Descriptive Set Theory

MATH 592 · Scribed Course Notes · Part 2: Definable Subsets of Polish Spaces

Definitions Results
Examples Notes
green text marks a term or notation at the point where it is defined

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Course Overview

Professor: Anush Tserunyan
Scribe: Yitian Zhang
Email: yit dot zha3 at gmail dot com

Note 1. The abbreviations used in the manuscript have been expanded in this edition.

Note 2. Text set in blue (or red) reproduces the scribe's own annotations, written in blue (or red) ink in the manuscript.

Note 3. The marker "Source manuscript page n" indicates where page n of the handwritten manuscript (Part 2) begins.

Note 4. Following the colour convention of the lecturer's notes, a term or a piece of notation is set in green at the point where it is defined, e.g. "a set is analytic if ..." or "$\mathcal{B}(X) := \dots$". Italics are reserved for ordinary emphasis.

Note 5. Results are numbered within sections, and a reference such as Theorem 2.1 is a clickable link to the result it cites; the number is set in this colour so that links are recognizable. Section references like §1.4 are links too.

Note 6. Results from Part 1 are cited by their number in the Part 1 edition of these notes, e.g. Part 1, Theorem 1.9 (Alexandrov's theorem). These are links into the file `Math592_Part1_Polish_Spaces.pdf`, which work when the two PDF files are kept in the same folder. Homework problems ("HW 3, Q2") are not part of the notes and are cited by name only.

Note 7. The manuscript ends (p. 78) with the heading of the next chapter, "Regularity properties of projective sets", which is not yet written and hence not transcribed.

Relevant readings as prerequisites.

- Analysis 1: § Topology and definitions.
- Analysis 2: § Metric spaces.
- Folland, *Real Analysis*, Chapter 4.

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Introduction. In Part 1, we discussed the properties of Polish spaces and their characterizations. In this part, we are mainly concerned with the properties of subsets of Polish spaces, and of functions between Polish spaces, putting our focus on the so-called "definability".

We begin our discussion with Borel measurability and Borel (measurable) sets and functions.

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1 σ-Algebras and Borel Sets

1.1 σ-Algebras, Measurable Spaces and Borel Sets

Definition 1.1

Recall that an algebra on a given set X is a collection of subsets $\mathcal{A} \subseteq \mathcal{P}(X)$ which contains $\varnothing$ and is closed under complements and finite unions (hence also finite intersections). A σ-algebra is an algebra which is closed under countable unions (hence also countable intersections). □

Definition 1.2 (Generated σ-algebra; dual collection)

Arbitrary intersections of σ-algebras on X are still σ-algebras, so for a given collection $\mathcal{E} \subseteq \mathcal{P}(X)$, it makes sense to define the σ-algebra generated by $\mathcal{E}$ as the intersection of all σ-algebras containing $\mathcal{E}$. In other words, this is the $\subseteq$-smallest σ-algebra containing $\mathcal{E}$, denoted $\langle \mathcal{E} \rangle_\sigma$, and $\langle \mathcal{E} \rangle$ denotes the algebra generated by $\mathcal{E}$.

For $\mathcal{E} \subseteq \mathcal{P}(X)$, denote $\neg\mathcal{E} := \{X \setminus A : A \in \mathcal{E}\}$, the dual collection of $\mathcal{E}$. □

Proposition 1.3

Let X be a set and $\varnothing \in \mathcal{E} \subseteq \mathcal{P}(X)$. Then $\langle \mathcal{E} \rangle_\sigma$ is the $\subseteq$-smallest collection $\mathcal{S}$ which contains $\mathcal{E} \sqcup \neg\mathcal{E}$ and is closed under countable unions and countable intersections.

Proof. Put $\widetilde{\mathcal{S}} := \{A \in \mathcal{S} : A, A^c \in \mathcal{S}\} \subseteq \mathcal{S}$. We show $\widetilde{\mathcal{S}}$ is a σ-algebra containing $\mathcal{E}$, hence $\widetilde{\mathcal{S}} \supseteq \langle \mathcal{E} \rangle_\sigma$, and $\widetilde{\mathcal{S}} \subseteq \mathcal{S} \subseteq \langle \mathcal{E} \rangle_\sigma$ (the latter since $\langle \mathcal{E} \rangle_\sigma$ is itself a collection with the two properties, and $\mathcal{S}$ is the smallest one), hence $\widetilde{\mathcal{S}} = \mathcal{S} = \langle \mathcal{E} \rangle_\sigma$.

Observe that $\mathcal{E} \subseteq \widetilde{\mathcal{S}}$ since $A \in \mathcal{E} \implies A^c \in \neg\mathcal{E}$, and $\mathcal{S} \supseteq \mathcal{E} \sqcup \neg\mathcal{E} \implies A, A^c \in \mathcal{S} \implies A \in \widetilde{\mathcal{S}}$. Moreover, $\widetilde{\mathcal{S}}$ is a σ-algebra: For $A \in \widetilde{\mathcal{S}}$, $A^c \in \widetilde{\mathcal{S}}$ by definition, and

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for $(A_n)_{n\in\mathbb{N}} \subseteq \widetilde{\mathcal{S}}$, $\bigcup_{n\in\mathbb{N}} A_n \in \mathcal{S}$ since $\mathcal{S}$ is closed under countable unions, and $\left(\bigcup_{n\in\mathbb{N}} A_n\right)^c = \bigcap_{n\in\mathbb{N}} A_n^c \in \mathcal{S}$ since $\mathcal{S}$ is closed under countable intersections; thus $\bigcup_{n\in\mathbb{N}} A_n \in \widetilde{\mathcal{S}}$. Lastly $\varnothing \in \widetilde{\mathcal{S}}$. Hence $\mathcal{E} \subseteq \widetilde{\mathcal{S}} \implies \langle \mathcal{E} \rangle_\sigma \subseteq \widetilde{\mathcal{S}}$. □

Definition 1.4

A measurable space is a set X equipped with a σ-algebra $\mathcal{I}$, i.e. a tuple $(X, \mathcal{I})$. For measurable spaces $(X, \mathcal{I})$, $(Y, \mathcal{J})$, a function $f : X \to Y$ is called measurable or $(\mathcal{I}, \mathcal{J})$-measurable if f^{-1} takes sets in $\mathcal{J}$ to sets in $\mathcal{I}$, i.e. $f^{-1}(B) \in \mathcal{I}$, $\forall B \in \mathcal{J}$.

When X is a topological space, the σ-algebra generated by the collection of all open sets is called the Borel σ-algebra of X, denoted $\mathcal{B}(X)$, and the sets in $\mathcal{B}(X)$ are called Borel sets.

For a function $f : X \to Y$ from a measurable space $(X, \mathcal{I})$ to a topological space Y, we say that f is measurable or $\mathcal{I}$-measurable if f is $(\mathcal{I}, \mathcal{B}(Y))$-measurable. Lastly, if X, Y are topological spaces, then a function $f : X \to Y$ is called Borel if it is $(\mathcal{B}(X), \mathcal{B}(Y))$-measurable. □

Note. If we ask for $(\mathcal{I},\mathcal{J})$-measurability instead of $(\mathcal{I},\mathcal{B}(Y))$-measurability (for a σ-algebra $\mathcal{J}$ on Y larger than $\mathcal{B}(Y)$), then the class of functions will be much smaller and less useful. □

Proposition 1.5

Continuous functions are Borel. More generally, let

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$(X,\mathcal{I})$, $(Y,\mathcal{J})$ be measurable spaces, $\mathcal{E}\subseteq\mathcal{J}$ generating $\mathcal{J}$, i.e. $\langle\mathcal{E}\rangle_\sigma=\mathcal{J}$. Then if $f:X\to Y$ satisfies $f^{-1}(B)\in\mathcal{I}$, $\forall B\in\mathcal{E}$, then f is $(\mathcal{I},\mathcal{J})$-measurable.

Proof. Note that the second part implies the first part. Define $\mathcal{J}_0:=\{B\in\mathcal{J}:f^{-1}(B)\in\mathcal{I}\}$. Then $\mathcal{J}_0\subseteq\mathcal{J}$ while $\mathcal{J}_0\supseteq\mathcal{E}$. Furthermore, $\mathcal{J}_0$ is a σ-algebra since $\varnothing=f^{-1}(\varnothing)\in\mathcal{I}$, and for $B,B_0,B_1,\ldots\in\mathcal{J}_0$, $f^{-1}(B^c)=f^{-1}(B)^c\in\mathcal{I}$, $f^{-1}(\bigcup_n B_n)=\bigcup_n f^{-1}(B_n)\in\mathcal{I}$, i.e. f^{-1} commutes with these operations and $\mathcal{I}$ is closed under them. Hence $\mathcal{J}_0=\mathcal{J}=\langle\mathcal{E}\rangle_\sigma$. □

1.2 Stratification of Borel Sets into a Hierarchy

Definition 1.6

Let X be a topological space. We define the hierarchy of Borel sets level-by-level, via transfinite recursion. Let ω_1 denote the first uncountable ordinal. For each $1\le\xi<\omega_1$, define the collections of subsets of X as follows:

$$\mathbf{\Sigma}^0_1(X):=\{U\subseteq X:U\text{ open}\},\qquad \mathbf{\Pi}^0_\xi(X):=\neg\mathbf{\Sigma}^0_\xi(X),$$
$$\mathbf{\Sigma}^0_\xi(X):=\Big\{\bigcup_{n\in\mathbb{N}}A_n:A_n\in\mathbf{\Pi}^0_{\xi_n}(X),\ \xi_n<\xi,\ \forall n\in\mathbb{N}\Big\}\quad(2\le\xi<\omega_1).$$

In addition, we define the so-called ambiguous classes:

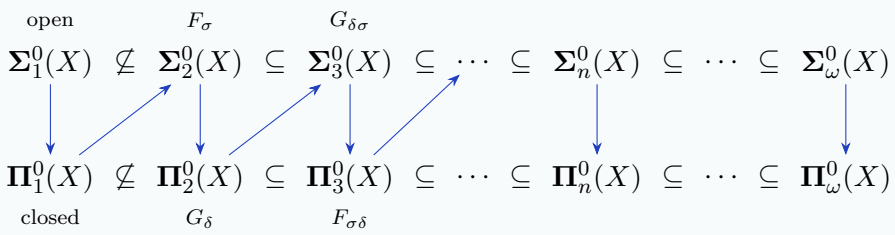
$$\mathbf{\Delta}^0_\xi(X):=\mathbf{\Sigma}^0_\xi(X)\cap\mathbf{\Pi}^0_\xi(X).\quad\square$$

Observation 1.7 (Monotonicity of the hierarchy)

Notice that by definition, for $2\le\xi<\eta$, $\mathbf{\Sigma}^0_\xi(X)\subseteq\mathbf{\Sigma}^0_\eta(X)$, since the collection we can pick the A_n's from is enlarged from ξ to η, and $\mathbf{\Pi}^0_\xi(X)=\neg\mathbf{\Sigma}^0_\xi(X)\subseteq\neg\mathbf{\Sigma}^0_\eta(X)=\mathbf{\Pi}^0_\eta(X)$.

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Hence we have the following diagram:



(Blue arrows: each $\mathbf{\Pi}^0_\xi$ is the dual of $\mathbf{\Sigma}^0_\xi$, and $\mathbf{\Sigma}^0_{\xi+1}$ consists of the countable unions of $\mathbf{\Pi}^0_\xi$ sets.) Here ω denotes the first infinite ordinal, i.e. $\mathbb{N}$ (ordered as usual). $\mathbf{\Sigma}^0_\omega(X)$ contains unions $\bigcup_{n\in\mathbb{N}} A_n$ where $A_n \in \mathbf{\Pi}^0_n(X)$, and there is no reason why $\bigcup_{n\in\mathbb{N}} A_n$ should appear in $\mathbf{\Sigma}^0_m(X)$, for some $m \in \mathbb{N}$. Thus to obtain the full Borel sets, we must continue transfinitely via the definition above (up to ω_1, as we will see).

In particular, $\mathbf{\Sigma}^0_1(X) \subseteq \mathbf{\Sigma}^0_2(X)$ is not true in general topological spaces, as open sets are not necessarily F_σ; symmetrically, $\mathbf{\Pi}^0_1(X) \subseteq \mathbf{\Pi}^0_2(X)$ is not true in general, as closed sets are not necessarily G_δ. □ □

Proposition 1.8

Let X be a metrizable space.

(a) $\mathbf{\Sigma}^0_\eta(X) \cup \mathbf{\Pi}^0_\eta(X) \subseteq \mathbf{\Delta}^0_\xi(X)$, $\forall\, 1 \le \eta < \xi < \omega_1$.
(b) $\mathcal{B}(X) = \bigcup_{1\le\xi<\omega_1} \mathbf{\Sigma}^0_\xi(X) = \bigcup_{1\le\xi<\omega_1} \mathbf{\Pi}^0_\xi(X) = \bigcup_{1\le\xi<\omega_1} \mathbf{\Delta}^0_\xi(X)$.

Proof. (a) Take $A \in \mathbf{\Sigma}^0_\eta(X)$. Then $A \in \mathbf{\Sigma}^0_\xi(X)$ by our observation above. In particular, $\mathbf{\Sigma}^0_1(X) \subseteq \mathbf{\Sigma}^0_2(X)$ since open sets are F_σ in metrizable spaces (lecture 1, Part 1, Lemma 1.8). $A \in \mathbf{\Pi}^0_\xi(X)$ by taking $A_0 = A \in \mathbf{\Sigma}^0_\eta(X)$, $A_n = X \in \mathbf{\Sigma}^0_1(X)$ for $n \ge 1$; then by definition, $A = \bigcap_{n\in\mathbb{N}} A_n \in \mathbf{\Pi}^0_\xi(X)$.

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Hence $A \in \mathbf{\Sigma}^0_\xi(X) \cap \mathbf{\Pi}^0_\xi(X) = \mathbf{\Delta}^0_\xi(X)$. Lastly, the dual simply takes the complement of the sets in the collection, thus

$$\begin{aligned}\mathbf{\Pi}^0_\eta(X) = \neg\mathbf{\Sigma}^0_\eta(X) &\subseteq \neg(\mathbf{\Sigma}^0_\xi(X) \cap \mathbf{\Pi}^0_\xi(X)) \\ &= (\neg\mathbf{\Sigma}^0_\xi(X)) \cap (\neg\mathbf{\Pi}^0_\xi(X)) = \mathbf{\Pi}^0_\xi(X) \cap \mathbf{\Sigma}^0_\xi(X) \\ &= \mathbf{\Delta}^0_\xi(X).\end{aligned}$$

Hence $\mathbf{\Sigma}^0_\eta(X) \cup \mathbf{\Pi}^0_\eta(X) \subseteq \mathbf{\Delta}^0_\xi(X)$.

(b) Notice that (a) implies that we have an intertwined chain of classes, and consequently

$$\bigcup_{\xi<\omega_1} \mathbf{\Sigma}^0_\xi(X) = \bigcup_{\xi<\omega_1} \mathbf{\Pi}^0_\xi(X) = \bigcup_{\xi<\omega_1} \mathbf{\Delta}^0_\xi(X) :$$

By definition of $\mathbf{\Delta}^0_\xi(X)$, $\bigcup_{\xi<\omega_1} \mathbf{\Sigma}^0_\xi(X) \supseteq \bigcup_{\xi<\omega_1} \mathbf{\Delta}^0_\xi(X)$ and $\bigcup_{\xi<\omega_1} \mathbf{\Pi}^0_\xi(X) \supseteq \bigcup_{\xi<\omega_1} \mathbf{\Delta}^0_\xi(X)$. On the other hand, by (a),

$$\begin{aligned}\bigcup_{\xi<\omega_1} \mathbf{\Delta}^0_\xi(X) \supseteq \bigcup_{1<\xi<\omega_1} \mathbf{\Delta}^0_\xi(X) &\supseteq \bigcup_{\eta<\omega_1} (\mathbf{\Sigma}^0_\eta(X) \cup \mathbf{\Pi}^0_\eta(X)) \\ &= \bigcup_{\xi<\omega_1} \mathbf{\Sigma}^0_\xi(X) \;\cup\; \bigcup_{\xi<\omega_1} \mathbf{\Pi}^0_\xi(X).\end{aligned}$$

It is left to show that $\mathcal{B}(X) =$ any of the unions. Clearly $\mathbf{\Sigma}^0_\xi(X) \subseteq \mathcal{B}(X)$ since at every inductive step we only take complements and countable unions. And $\mathbf{\Sigma}^0_1(X)$ generates $\mathcal{B}(X)$. Hence it remains to show the unions are σ-algebras. Indeed, closure under complement follows from $\neg\mathbf{\Delta}^0_\xi(X) = \mathbf{\Delta}^0_\xi(X)$,

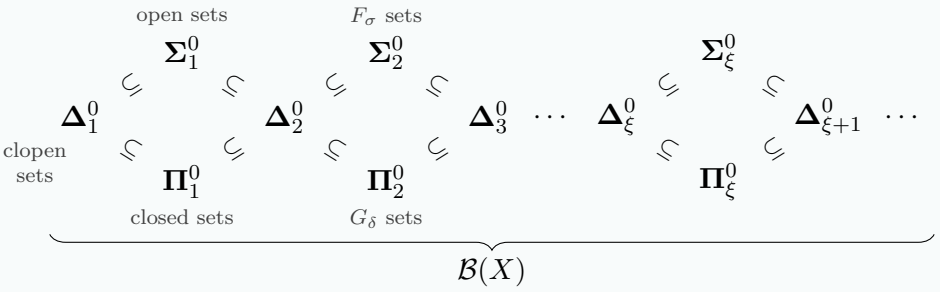
and closure under countable union follows from each $\mathbf{\Sigma}^0_\xi(X)$ being closed under countable union (each set in $\mathbf{\Sigma}^0_\xi(X)$ is a countable union of $A_n \in \mathbf{\Pi}^0_{\xi_n}(X)$,

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and their countable union is a countable union of countable unions, hence still a countable union), together with the fact that countably many sets $A_n \in \mathbf{\Sigma}^0_{\xi_n}(X)$ all lie in one $\mathbf{\Sigma}^0_\xi(X)$, namely for $\xi := \sup_n \xi_n < \omega_1$ (a countable supremum of countable ordinals is countable). □

Observation 1.9 (The Borel hierarchy of a metrizable space)

Thus for metrizable spaces, we have the following picture:



From HW 1, every second countable space has $\leq |2^{\mathbb{N}}| = \mathfrak{c}$ many open sets, hence also closed sets. Taking countable unions of sets from a collection of cardinality $\leq \mathfrak{c}$ still results in $\leq |2^{\mathbb{N}}|$ many sets since $(2^{\mathbb{N}})^{\mathbb{N}} \cong 2^{\mathbb{N}\times\mathbb{N}} \cong 2^{\mathbb{N}}$. Thus by transfinite induction on ξ, $|\mathbf{\Sigma}^0_\xi(X)| \leq \mathfrak{c}$ and $|\mathbf{\Pi}^0_\xi(X)| \leq \mathfrak{c}$. Finally

$$|\mathcal{B}(X)| = \Big|\bigcup_{1\leq\xi<\omega_1} \mathbf{\Sigma}^0_\xi(X)\Big| \leq |\omega_1| \times |2^{\mathbb{N}}| \leq |2^{\mathbb{N}} \times 2^{\mathbb{N}}| = |2^{\mathbb{N}}|,$$

with the last equality given by the interleaving map $((a_n)_n, (b_n)_n) \mapsto (a_1, b_1, a_2, b_2, \dots)$. Hence there are at most continuum-many Borel sets in a second countable space, and continuum is preserved along the zigzag chain above. In contrast, $\mathcal{L}$:= the Lebesgue measurable subsets of $\mathbb{R}$ has cardinality at least $2^{\mathfrak{c}} = 2^{|2^{\mathbb{N}}|}$,

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since the $\frac{1}{3}$-Cantor set has cardinality continuum with Lebesgue measure 0, thus all of its subsets are Lebesgue measurable with measure 0. □

Proposition 1.10

Let X be a topological space and $1 \leq \xi < \omega_1$, i.e. ξ a countable ordinal.

(a) $\mathbf{\Sigma}^0_\xi(X)$ is closed under taking countable unions; $\mathbf{\Pi}^0_\xi(X)$ is closed under taking countable intersections; $\mathbf{\Delta}^0_\xi(X)$ is closed under taking complements.
(b) If X is metrizable, then all three classes are closed under taking finite unions and intersections.

Proof. (a) $\mathbf{\Sigma}^0_\xi(X)$ is closed under taking countable unions since for $\xi = 1$, a countable union of open sets is still open; for $\xi \geq 2$, a countable union of countable unions is still a countable union. $\mathbf{\Pi}^0_\xi(X)$

follows by taking complements with $\mathbf{\Sigma}^0_\xi(X)$, and

$$\neg\mathbf{\Delta}^0_\xi(X) = \neg(\mathbf{\Sigma}^0_\xi(X) \cap \mathbf{\Pi}^0_\xi(X)) = \neg\mathbf{\Sigma}^0_\xi(X) \cap \neg\mathbf{\Pi}^0_\xi(X) = \mathbf{\Pi}^0_\xi(X) \cap \mathbf{\Sigma}^0_\xi(X) = \mathbf{\Delta}^0_\xi(X).$$

(b) It suffices to show $\mathbf{\Sigma}^0_\xi(X)$ is closed under finite intersections and then everything else follows. Take $\bigcup_{n\in\mathbb{N}} A_n, \bigcup_{n\in\mathbb{N}} B_n \in \mathbf{\Sigma}^0_\xi(X)$. Then $(\bigcup_{n\in\mathbb{N}} A_n) \cap (\bigcup_{n\in\mathbb{N}} B_n) = \bigcup_{n,m\in\mathbb{N}}(A_n \cap B_m)$, where $A_n \in \mathbf{\Pi}^0_{\xi_n}(X)$, $B_m \in \mathbf{\Pi}^0_{\eta_m}(X)$, $\xi_n, \eta_m < \xi$. Hence $A_n, B_m \in \mathbf{\Pi}^0_\eta(X)$, $\eta = \max(\xi_n, \eta_m) < \xi$, since $\mathbf{\Pi}^0_\zeta(X) \subseteq \mathbf{\Pi}^0_\eta(X)$ for $\zeta < \eta$ (Proposition 1.8(a)), and $A_n \cap B_m \in \mathbf{\Pi}^0_\eta(X)$ by (a). Thus $\bigcup_{n,m\in\mathbb{N}}(A_n \cap B_m) \in \mathbf{\Sigma}^0_\xi(X)$ by definition. □

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Example 1.11

Recall we proved for X a compact metric space, Y a separable complete metric space, that $C(X,Y)$ is Polish with respect to the uniform metric (HW 1). We show the set of continuously differentiable functions on $[0,1]$, $C^1([0,1])$, is $\mathbf{\Pi}^0_3$ as a subset of $C([0,1]) = C([0,1],\mathbb{R})$, i.e. $C^1([0,1]) \in \mathbf{\Pi}^0_3(C([0,1]))$, i.e. having a definition of the form $\forall^{\mathbb{N}}\exists^{\mathbb{N}}\forall^{\mathbb{N}}$.

We would reduce the complexity of the definition by reducing the complexity of quantifiers (e.g. $\forall\varepsilon \in \mathbb{R}^+ \rightsquigarrow \forall\varepsilon \in \mathbb{Q}^+$) and applying theorems.

For $f \in C([0,1])$, $f \in C^1([0,1]) \iff \forall\varepsilon \in \mathbb{Q}^+$, $\exists$ finitely many rational open intervals $I_0, \dots, I_{n-1}$ covering $[0,1]$ such that $\forall\, 0 \le j < n$, $f \in A_{I_j,\varepsilon}$, where

$$A_{I_j,\varepsilon} := \left\{ f \in C([0,1]) : \forall\, a \neq b,\ c \neq d \text{ in } I_j,\ \left|\frac{f(a)-f(b)}{a-b} - \frac{f(c)-f(d)}{c-d}\right| \le \varepsilon \right\}. \qquad (*)$$

[Exercise. Hint: Recall that $g(x)$ is continuous at x_0 iff for all $x_n \xrightarrow{n\to\infty} x_0$, $g(x_n) \to g(x_0)$, and noticing that $(*)$ implies Cauchyness and convergence of g at once, we get differentiability at $x \in [0,1]$. Moreover, $(*)$ implies continuity of f' as well.]

Now fixing I_j with $a,b,c,d \in I_j$, the map

$$F_{a,b,c,d} : C([0,1]) \to [0,\infty), \qquad f \mapsto \left|\frac{f(a)-f(b)}{a-b} - \frac{f(c)-f(d)}{c-d}\right|$$

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is easily continuous since every $f \in C([0,1])$ is uniformly continuous. Then

$$A_{I_j,\varepsilon} = \underbrace{\bigcap_{\substack{a,b,c,d\in I_j \\ a\neq b,\ c\neq d}} \underbrace{F^{-1}_{a,b,c,d}([0,\varepsilon])}_{\text{continuous preimage of closed sets = closed}}}_{\bigcap \text{ closed=closed}}$$

is closed in $C([0,1])$. Thus

$$f \in C^1([0,1]) \iff \underbrace{\forall\varepsilon \in \mathbb{Q}^+,\ \underbrace{\exists\,(I_0,\dots,I_{n-1}) \in \text{rational open intervals}^{<\mathbb{N}},\ \underbrace{\forall j < n,\ f \in A_{I_j,\varepsilon}}_{\mathbf{\Pi}^0_1=\text{closed}}}_{\mathbf{\Sigma}^0_2=F_\sigma}}_{\mathbf{\Pi}^0_3=\neg\mathbf{\Sigma}^0_3}$$

(Alternatively, $\mathbf{\Pi}^0_3$ is closed under countable intersections and $\mathbf{\Sigma}^0_2 \subseteq \mathbf{\Delta}^0_3 \subseteq \mathbf{\Pi}^0_3$.) Explicitly, if $\mathcal{I} := \{(p,q) \cap [0,1] : p < q \in \mathbb{Q}\}$, which is countable (! $\mathbb{N}^{<\mathbb{N}}$ is countable, $\mathbb{N}^{\mathbb{N}}$ is uncountable, $= |2^{\mathbb{N}}| = \mathfrak{c}$), then

$$C^1([0,1]) = \bigcap_{\varepsilon \in \mathbb{Q}^+} \bigcup_{\substack{(I_0, I_1, \ldots, I_{n-1}) \in \mathcal{I}^{<\mathbb{N}} \\ \bigcup_j I_j = [0,1]}} \bigcap_{0 \le j < n} A_{I_j, \varepsilon}. \quad \square$$

1.3 The Classes $\mathcal{B}$, $\mathbf{\Sigma}^0_\xi$, $\mathbf{\Pi}^0_\xi$, $\mathbf{\Delta}^0_\xi$

Definition 1.12

By $\mathcal{B}$, $\mathbf{\Sigma}^0_\xi$, $\mathbf{\Pi}^0_\xi$, $\mathbf{\Delta}^0_\xi$, we denote the class of the corresponding sets in all topological spaces, and for each such space X, we write $\mathcal{B}(X)$, $\mathbf{\Sigma}^0_\xi(X)$, $\mathbf{\Pi}^0_\xi(X)$, $\mathbf{\Delta}^0_\xi(X)$ to mean the corresponding set of subsets of X. $\square$

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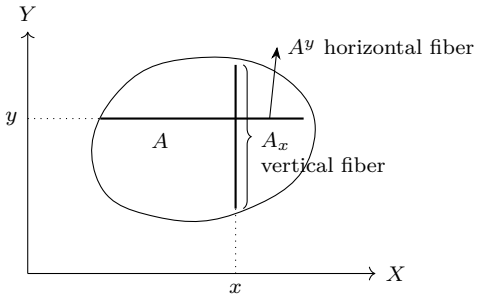
Notation. For a set $A \subseteq X \times Y$, we denote by

$$A_x := \{y \in Y : (x,y) \in A\}, \quad \text{for } x \in X, \qquad \text{and} \qquad A^y := \{x \in X : (x,y) \in A\}, \quad \text{for } y \in Y,$$

the vertical and horizontal fibers over the point $x \in X$ and $y \in Y$ respectively. Similarly, for a function $f : X \times Y \to Z$, denote by

$$f_x : Y \to Z,\ y \mapsto f(x,y) \qquad \text{and} \qquad f^y : X \to Z,\ x \mapsto f(x,y)$$

the vertical and horizontal fibers over the point $x \in X$ and $y \in Y$, respectively. $\square$



Proposition 1.13

Let Γ be any one of the classes $\mathbf{\Sigma}^0_\xi$, $\mathbf{\Pi}^0_\xi$, $\mathbf{\Delta}^0_\xi$, $\xi < \omega_1$. Then Γ is closed under

(a) continuous preimages, i.e. if X, Y are topological spaces, $f : X \to Y$ is continuous and $B \in \Gamma(Y)$, then $f^{-1}(B) \in \Gamma(X)$;

(b) putting countably many sets as fibers over $\mathbb{N}$, i.e. if X is a topological space, $A_n \in \Gamma(X)$, then the set $B \subseteq \mathbb{N} \times X$ defined by $B_n := A_n$ is in $\Gamma(\mathbb{N} \times X)$, where B_n is the fiber of B over $n \in \mathbb{N}$, $\mathbb{N} \times X$ is equipped with the product topology, and $\mathbb{N}$ is equipped with the discrete topology.

Proof. (a) Note that the statement with $\mathbf{\Delta}^0_\xi$ follows from $\mathbf{\Sigma}^0_\xi$ and $\mathbf{\Pi}^0_\xi$ from the fact that taking intersections commutes with taking preimages. We show the statement for $\mathbf{\Sigma}^0_\xi$ and $\mathbf{\Pi}^0_\xi$ simultaneously by transfinite induction on ξ.

Base case: $\xi = 1$. Continuous preimages of open sets ($\mathbf{\Sigma}^0_1(Y)$) are

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open ($\mathbf{\Sigma}^0_1(X)$). $\mathbf{\Pi}^0_1$ follows by taking the dual collection.

Inductive step: Our definition of $\mathbf{\Sigma}^0_\xi$ and $\mathbf{\Pi}^0_\xi$ does not split into successor / limit cases, thus we need not split into cases in our transfinite induction proof. Suppose $\mathbf{\Sigma}^0_\eta$ and $\mathbf{\Pi}^0_\eta$ are closed under continuous preimages, for all $\eta < \xi$. Then for $\bigcup_n A_n \in \mathbf{\Sigma}^0_\xi(Y)$, with $A_n \in \mathbf{\Pi}^0_{\xi_n}(Y)$, $\xi_n < \xi$, $\forall n$, we have $f^{-1}(\bigcup_n A_n) = \bigcup_n f^{-1}(A_n)$ with each $f^{-1}(A_n) \in \mathbf{\Pi}^0_{\xi_n}(X)$, $\xi_n < \xi$, by the induction hypothesis. Thus by definition, $f^{-1}(\bigcup_n A_n) = \bigcup_n f^{-1}(A_n) \in \mathbf{\Sigma}^0_\xi(X)$. (And $\mathbf{\Pi}^0_\xi$ follows by taking the dual collection.)

(b) Induct transfinitely on ξ in the same way as above. Alternatively, consider $\pi : \mathbb{N} \times X \to X$ the projection map, which is continuous by definition of the product topology. Then for $\bigcup_n A_n \in \mathbf{\Sigma}^0_\xi(X)$,

$$B = \bigcup_n (\{n\} \times B_n) = \bigcup_n \left[(\{n\} \times X) \cap \pi^{-1}(A_n) \right]$$

is a countable union of (clopen ($\mathbf{\Delta}^0_1(X) \subseteq \mathbf{\Sigma}^0_\xi(X)$) sets) $\cap$ ($\mathbf{\Sigma}^0_\xi(\mathbb{N} \times X)$ sets). We show that (clopen set) $\cap$ ($\mathbf{\Sigma}^0_\xi$ set) is $\mathbf{\Sigma}^0_\xi$. Indeed, clopen sets are $\mathbf{\Delta}^0_1$, hence $\mathbf{\Pi}^0_1$ and $\mathbf{\Sigma}^0_1$, hence $\mathbf{\Pi}^0_2$ since $\mathbf{\Pi}^0_1 \subseteq \mathbf{\Sigma}^0_2 \implies \mathbf{\Sigma}^0_1 = \neg\mathbf{\Pi}^0_1 \subseteq \neg\mathbf{\Sigma}^0_2 = \mathbf{\Pi}^0_2$, hence $\mathbf{\Pi}^0_\eta$ $\forall\, \eta < \omega_1$ since $\mathbf{\Pi}^0_2 \subseteq \mathbf{\Pi}^0_3 \subseteq \cdots$ in any topological space. Let C be clopen and $\bigcup_n D_n \in \mathbf{\Sigma}^0_\xi$ with $D_n \in \mathbf{\Pi}^0_{\xi_n}$, $\xi_n < \xi$ $\forall n$. Then $C \cap (\bigcup_n D_n) = \bigcup_n (C \cap D_n)$, where each $C \cap D_n$ is an intersection in $\mathbf{\Pi}^0_{\xi_n}$. Hence $C \cap D_n \in \mathbf{\Pi}^0_{\xi_n}$ since the $\mathbf{\Pi}^0_\eta$ classes are closed under countable intersections. Thus $\bigcup_n (C \cap D_n) \in \mathbf{\Sigma}^0_\xi$. Hence $B = \bigcup_n \left[(\{n\} \times X) \cap \pi^{-1}(A_n) \right] \in \mathbf{\Sigma}^0_\xi(\mathbb{N} \times X)$.

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For $\mathbf{\Pi}^0_\xi$, notice that $(m, x) \in B = \bigcup_n \{n\} \times B_n \iff \forall n \in \mathbb{N}$, $n \neq m$ or $x \in A_n$, thus

$$B = \underbrace{\bigcap_{n \in \mathbb{N}} \Big[\underbrace{\underbrace{((\mathbb{N} \setminus \{n\}) \times X)}_{\mathbf{\Delta}^0_1 \subseteq \mathbf{\Pi}^0_\xi} \cup \underbrace{\pi^{-1}(A_n)}_{\mathbf{\Pi}^0_\xi}}_{\mathbf{\Pi}^0_\xi \ (*)} \Big]}_{\mathbf{\Pi}^0_\xi \text{ (by the closure property)}}.$$

To show $(*)$, first show that every set in $\mathbf{\Pi}^0_\xi$ takes the form $\bigcap_n E_n$ with $E_n \in \mathbf{\Sigma}^0_{\xi_n}$, $\xi_n < \xi$, then rewrite the union term as some countable $\bigcap$ in $\mathbf{\Pi}^0_\xi$ (distribute the clopen set into the intersection; a clopen set union a $\mathbf{\Sigma}^0_{\xi_n}$ set is $\mathbf{\Sigma}^0_{\xi_n}$ by the dual of what we just showed). □

Definition 1.14

Let Γ be one of the classes. Denote the dual class by $\neg\Gamma$, i.e. if X is a topological space, then

$$\neg\Gamma(X) := \{A \subseteq X : A^c \in \Gamma(X)\}, \qquad \text{thus } \mathbf{\Sigma}^0_\xi = \neg\mathbf{\Pi}^0_\xi.$$

Furthermore, for topological spaces X, Y and $A \subseteq X \times Y$, denote $\text{proj}_X(A) := \{x \in X : \exists\, y \in$

Y such that $(x,y)\in A\}$, and put

$$\exists^Y\Gamma(X) := \{\mathrm{proj}_X(A) : A\in\Gamma(X\times Y)\}\subseteq\mathcal{P}(X),$$
$$\forall^Y\Gamma(X) := \neg\exists^Y\neg\Gamma(X) := \neg(\exists^Y(\neg\Gamma)(X))\subseteq\mathcal{P}(X).$$

I.e.,

$$\begin{aligned}
S\in\exists^Y\Gamma(X) &\iff \exists\, A\in\Gamma(X\times Y)\ \forall x\in X,\ \big[x\in S \iff \exists\, y\in Y\ (x,y)\in A\big],\\
S\in\forall^Y\Gamma(X) &\iff S^c\in(\exists^Y(\neg\Gamma))(X)\\
&\iff \exists\, A\in\neg\Gamma(X\times Y)\ \forall x\in X\ \big[x\in S^c \iff \exists\, y\in Y\ (x,y)\in A\big]\\
&\iff \exists\, C\in\Gamma(X\times Y)\ \forall x\in X\ \big[x\in S \iff \forall y\in Y\ (x,y)\notin A\big]\quad (C=A^c)\\
&\iff \exists\, C\in\Gamma(X\times Y)\ \forall x\in X\ \big[x\in S \iff \forall y\in Y\ (x,y)\in C\big].
\end{aligned}$$

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By $\neg\exists^Y\neg\Gamma(X)$, we mean: apply the operator $\exists^Y$ to the class $\neg\Gamma$ first, then apply $\exists^Y\neg\Gamma$ to X, and finally take the dual collection $\neg(\exists^Y\neg\Gamma(X))$. □

Corollary 1.15

$\forall\, 1\le\xi<\omega_1$, for X metrizable, $\mathbf{\Sigma}^0_{\xi+1}(X)=\exists^{\mathbb{N}}\mathbf{\Pi}^0_\xi(X)$. Thus also $\mathbf{\Pi}^0_{\xi+1}(X)=\forall^{\mathbb{N}}\mathbf{\Sigma}^0_\xi(X)$.

Proof. $\exists^{\mathbb{N}}\mathbf{\Pi}^0_\xi(X)\subseteq\mathbf{\Sigma}^0_{\xi+1}(X)$: Take $S\in\exists^{\mathbb{N}}\mathbf{\Pi}^0_\xi(X)$, with $A\in\mathbf{\Pi}^0_\xi(\mathbb{N}\times X)$ such that $\mathrm{proj}_X(A)=S$. Then observe that $\mathrm{proj}_X(A)=\bigcup_{n\in\mathbb{N}}A_n$, with each fiber $A_n=$ the continuous preimage of A by $\iota_n: x\mapsto(n,x)$. By Proposition 1.13(a), each A_n is $\mathbf{\Pi}^0_\xi(X)$. Hence $S\in\mathbf{\Sigma}^0_{\xi+1}(X)$.

$\mathbf{\Sigma}^0_{\xi+1}(X)\subseteq\exists^{\mathbb{N}}\mathbf{\Pi}^0_\xi(X)$: Take $S=\bigcup_{n\in\mathbb{N}}A_n$, with $A_n\in\mathbf{\Pi}^0_{\xi_n}(X)$, $\xi_n<\xi+1$, i.e. $\xi_n\le\xi$. Thus $A_n\in\mathbf{\Pi}^0_\xi(X)$, $\forall n$ (using metrizability so that $\mathbf{\Pi}^0_1\subseteq\mathbf{\Pi}^0_2$). By Proposition 1.13(b), A with the A_n as fibers over n is $\mathbf{\Pi}^0_\xi(\mathbb{N}\times X)$. Thus $S=\mathrm{proj}_X(A)$ is $\exists^{\mathbb{N}}\mathbf{\Pi}^0_\xi(X)$ by definition.

Lastly $\mathbf{\Pi}^0_{\xi+1}(X)=\neg\mathbf{\Sigma}^0_{\xi+1}(X)=\neg\exists^{\mathbb{N}}\mathbf{\Pi}^0_\xi(X)=\neg\exists^{\mathbb{N}}\neg\mathbf{\Sigma}^0_\xi(X)=\forall^{\mathbb{N}}\mathbf{\Sigma}^0_\xi(X)$. □

Remark 1.16

Thus $\bigcup_{1\le n<\omega}\mathbf{\Sigma}^0_n$ are all the sets obtained from open sets by $\neg$ and $\exists^{\mathbb{N}}$ (finite $\cap$ and $\cup$ can be encoded by $\neg$ and $\exists^{\mathbb{N}}$; in contrast, infinite $\cap$ and $\cup$ could escape to $\mathbf{\Sigma}^0_\omega$ outside of $\bigcup_{1\le n<\omega}\mathbf{\Sigma}^0_n$).

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Because of this, the superscript "0" refers to the fact that we quantify over $\mathbb{N}$ as opposed to $2^{\mathbb{N}}$, i.e. $\mathcal{P}(\mathbb{N})$. $\mathbf{\Sigma}^1_n$, $\mathbf{\Pi}^1_n$ would then be used when we allow quantification over $2^{\mathbb{N}}$ or $\mathbb{N}^{\mathbb{N}}$ or $\mathbb{R}$ or any uncountable Polish space. □

1.4 Universal Sets and Diagonalization

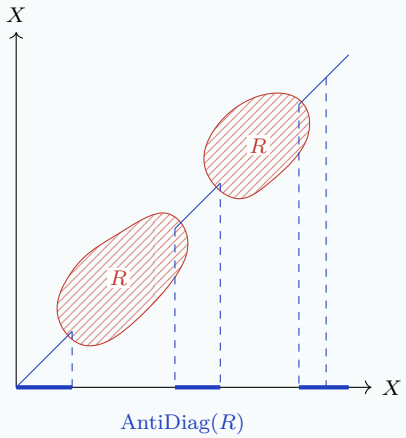
Aside. A very recent work done by Evan Leach (MIT, UCLA, Sep. 2026) presented a combinatorial proof using infinite circuits of what we are going to prove next using diagonalization (the second corollary below, Corollary 1.24).

Lemma 1.17 (Diagonalization Lemma (Cantor))

For a set X and $R \subseteq X \times X$, the antidiagonal set

$$\mathrm{AntiDiag}(R) := \{x \in X : (x, x) \notin R\}$$

is not a vertical or horizontal fiber of R, i.e. $\mathrm{AntiDiag}(R) \neq R_x$ and $\mathrm{AntiDiag}(R) \neq R^x$, $\forall x \in X$.



Proof. Notice that for every $x \in X$, $x \in \mathrm{AntiDiag}(R) \iff (x, x) \notin R \iff x \notin R_x$ (and likewise $\iff x \notin R^x$), thus $\mathrm{AntiDiag}(R) \neq R_x$ (and $\neq R^x$), for any $x \in X$. □

Corollary 1.18

$\mathbb{R}$ and $2^{\mathbb{N}}$ are uncountable.

We illustrate that the lemma above is really generalizing

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the elementary digit-flipping proof of $\mathbb{R}$'s uncountability.

Proof. Suppose $(x_n)_{n\in\mathbb{N}} \subseteq 2^{\mathbb{N}}$, i.e. each x_i is a 0-1 sequence. We can view it as $R \subseteq \mathbb{N} \times \mathbb{N}$ where $(m, n) \in R \iff x_m(n) = 1$. Hence the fiber $R_m = \{n : x_m(n) = 1\}$ is just the set of entries of the m-th sequence having a "1". Consider $\mathrm{AntiDiag}(R) = \{n : x_n(n) = 0\}$. Flipping the digits, we get $y(n) := 1 - x_n(n)$ (the characteristic function of $\mathrm{AntiDiag}(R)$). By Cantor's lemma, $\mathrm{AntiDiag}(R) \neq R_m$, $\forall m$, thus $y \neq x_m$ for any m. So $y \in 2^{\mathbb{N}} \setminus (x_n)_{n\in\mathbb{N}}$. One can inject $2^{\mathbb{N}} \hookrightarrow \mathbb{R}$ via the decimal representation of reals, hence $|\mathbb{N}| < |2^{\mathbb{N}}| \leq |\mathbb{R}|$. □

Exercise 1.19

Use the lemma and show $|X| < |\mathcal{P}(X)|$, i.e. no surjection $X \twoheadrightarrow \mathcal{P}(X)$ exists. (Hint: Suppose such $X \twoheadrightarrow \mathcal{P}(X)$ exists, then view it as a correspondence / subset of $X \times X$. What are the vertical fibres and what is the antidiagonal in this case? Correspondence = set-valued function.) □

Definition 1.20

Let Γ be a class of sets in topological spaces. Let X, Y be topological spaces. We say that a set $U \subseteq Y \times X$ parametrizes $\Gamma(X)$ if $\{U_y : y \in Y\} = \Gamma(X)$. Think of Y as a parameter set, and decoding along Y gives $\Gamma(X)$. If moreover U itself is in Γ, i.e. $U \in \Gamma(Y \times X)$, then we call U a Y-universal set for $\Gamma(X)$. □

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Observation 1.21 (Fibers are preimages)

Let Γ be any class of sets in topological spaces (i.e. Γ is an arbitrary class we can define over a topological space) which is closed under taking continuous preimages. Then Γ is closed under taking fibers, since if $R \in \Gamma(X \times Y)$ and $x \in X$, then R_x is the preimage of R under the continuous function $Y \to X \times Y$, $y \mapsto (x, y)$. □

Note. We have used this observation once already, when we proved Corollary 1.15.

Theorem 1.22

Let X be second countable metrizable. Then for $1 \leq \xi < \omega_1$, there is a $2^{\mathbb{N}}$-universal set for $\mathbf{\Sigma}^0_\xi(X)$ and for $\mathbf{\Pi}^0_\xi(X)$.

Proof. Transfinite induction on ξ.

$\xi = 1$. Fix a countable basis $(W_n)_{n\in\mathbb{N}}$ for X. Then define $U \subseteq 2^{\mathbb{N}} \times X$: for each $z \in 2^{\mathbb{N}}$,

$$U_z := \bigcup \{W_n : n \in \mathbb{N} \text{ and } z(n) = 1\}.$$

It is clear that $U = \bigcup_{z\in 2^{\mathbb{N}}} \{z\} \times U_z$ parametrizes $\mathbf{\Sigma}^0_1(X)$. Furthermore, U is $\mathbf{\Sigma}^0_1(2^{\mathbb{N}} \times X)$, i.e. open, because

$$\begin{aligned}(z, x) \in U &\iff x \in \bigcup \{W_n : n \in \mathbb{N},\ z(n) = 1\} \\ &\iff \underbrace{\underbrace{\exists n \in \mathbb{N}}_{\text{(countable) union}} \Big(\underbrace{\underbrace{z(n) = 1}_{\substack{\text{clopen} \\ (\{z : z(n)=1\} \times X) \\ \text{clopen (Part 1, trees)}}} \wedge \underbrace{x \in W_n}_{\substack{\text{open} \\ (2^{\mathbb{N}} \times W_n)}}}_{\text{open}} \Big)}_{\text{open}}\end{aligned}$$

(the cylinder sets $\{z : z(n) = 1\}$ are clopen by the note on cylinder sets in Part 1).

Inductive step. Suppose there is a $2^{\mathbb{N}}$-universal set $U^{(\eta)}$ for $\mathbf{\Sigma}^0_\eta(X)$, for each

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$\eta < \xi$. Then $V^{(\eta)} := \left(U^{(\eta)}\right)^c$ is a $2^{\mathbb{N}}$-universal set for $\mathbf{\Pi}^0_\eta(X)$ because fibers commute with complements from the definition (exercise). It remains to build a universal set for $\mathbf{\Sigma}^0_\xi(X)$.

Fix $\eta_n < \xi$ with $\eta_n \leq \eta_{n+1}$, $\forall n \in \mathbb{N}$, such that $\sup\{\eta_n + 1 : n \in \mathbb{N}\} = \xi$. Take $B = \bigcup_k C_k \in \mathbf{\Sigma}^0_\xi(X)$, with each $C_k \in \mathbf{\Pi}^0_{\zeta_k}$, $\zeta_k < \xi$, $\forall k$. We rewrite B as follows: for each $\zeta < \xi = \sup_n(\eta_n + 1)$, there must be some N with $\zeta \in \eta_N + 1$, i.e. $\zeta < \eta_N + 1$, i.e. $\zeta \leq \eta_N$. Then since the sequence is increasing, $\forall n \geq N$, $\eta_n \geq \eta_N \geq \zeta$. Now give each C_k its own slot: for each k, denote the N for ζ_k by N_k. Inductively set $n_k := \max(N_k, n_{k-1} + 1)$ so that $n_0 < n_1 < n_2 < \cdots$ and $\eta_{n_k} \geq \eta_{N_k} \geq \zeta_k$ from our choice of N. Put $B_{n_k} := C_k$, and for the rest, the unused slots, $B_n := \varnothing$. Observe that we rewrote $B = \bigcup_n B_n$ with $B_n \in \mathbf{\Pi}^0_{\eta_n}$, and particularly $B_{n_k} = C_k \in \mathbf{\Pi}^0_{\zeta_k} \subseteq \mathbf{\Pi}^0_{\eta_{n_k}}$.

Now to encode B, we need to encode all B_n into one code. We use $2^{\mathbb{N}} \cong (2^{\mathbb{N}})^{\mathbb{N}}$ (see the proof of Part 1, Theorem 3.11), so it's enough to show the existence of a $(2^{\mathbb{N}})^{\mathbb{N}}$-universal set for $\mathbf{\Sigma}^0_\xi(X)$. To this end, define $U \subseteq (2^{\mathbb{N}})^{\mathbb{N}} \times X$ by setting, for each $z \in (2^{\mathbb{N}})^{\mathbb{N}}$,

$$U_z := \bigcup_{n \in \mathbb{N}} \left(V^{(\eta_n)}\right)_{z(n)}.$$

From the definition, $U = \bigcup_{z \in (2^{\mathbb{N}})^{\mathbb{N}}} \{z\} \times U_z$ parametrizes $\mathbf{\Sigma}^0_\xi(X)$. Furthermore, U is itself

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in $\mathbf{\Sigma}^0_\xi((2^{\mathbb{N}})^{\mathbb{N}} \times X)$ since

$$(z,x) \in U \iff x \in \bigcup_{n \in \mathbb{N}} \left(V^{(\eta_n)}\right)_{z(n)} \iff \underbrace{\underbrace{\exists n \in \mathbb{N}}_{\text{countable } \cup}\,,\ \underbrace{(z(n),x) \in V^{(\eta_n)}}_{\mathbf{\Pi}^0_{\eta_n}}}_{\mathbf{\Sigma}^0_\xi}$$

(the set of (z,x) with $(z(n),x) \in V^{(\eta_n)}$ is the preimage of $V^{(\eta_n)}$ under the continuous map $(z,x) \mapsto (z(n),x)$, hence $\mathbf{\Pi}^0_{\eta_n}$ by Proposition 1.13(a)). Universality follows. □

Corollary 1.23

Let X be second countable metrizable, Y an uncountable Polish space. For each $1 \leq \xi < \omega_1$, there is a Y-universal set for $\mathbf{\Sigma}^0_\xi(X)$, as well as one for $\mathbf{\Pi}^0_\xi(X)$.

Proof. We build Y-universal sets for $\mathbf{\Pi}^0_\xi(X)$, then take complements to get Y-universal sets for $\mathbf{\Sigma}^0_\xi(X)$. By the perfect set property for Polish spaces (i.e. the Cantor–Bendixson theorem together with Part 1, Theorem 4.5), since Y is uncountable, there is a homeomorphic copy $C \subseteq Y$ of $2^{\mathbb{N}}$. Thus by Theorem 1.22, there is a C-universal set $U \in \mathbf{\Pi}^0_\xi(C \times X)$ for $\mathbf{\Pi}^0_\xi(X)$. Clearly U parametrizes $\mathbf{\Pi}^0_\xi(X)$; it remains to observe

$$\mathbf{\Pi}^0_\xi(C \times X) = \mathbf{\Pi}^0_\xi(Y \times X)\big|_{C \times X} := \{A \cap (C \times X) : A \in \mathbf{\Pi}^0_\xi(Y \times X)\}$$

(HW 3, Q2). But because C is closed, for any $A \in \mathbf{\Pi}^0_\xi(Y \times X)$, $A \cap (C \times X) \in \mathbf{\Pi}^0_\xi(Y \times X)$, since $C \times X$ is closed ($\mathbf{\Pi}^0_1(Y \times X)$), thus $\mathbf{\Pi}^0_\xi(Y \times X)$, and $\mathbf{\Pi}^0_\xi$ is closed under countable $\cap$. Hence $U \in \mathbf{\Pi}^0_\xi(C \times X) = \mathbf{\Pi}^0_\xi(Y \times X)|_{C \times X} \subseteq \mathbf{\Pi}^0_\xi(Y \times X)$, so U is Y-universal for $\mathbf{\Pi}^0_\xi(X)$. □

Corollary 1.24

For every uncountable Polish space X and every $1 \leq \xi < \omega_1$, $\mathbf{\Sigma}^0_\xi(X) \neq \mathbf{\Pi}^0_\xi(X)$. Consequently, we get the so-called strictness of the Borel hierarchy in uncountable Polish spaces,

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that

$$\mathbf{\Delta}^0_1(X) \subsetneq \begin{matrix}\mathbf{\Sigma}^0_1(X)\\ \mathbf{\Pi}^0_1(X)\end{matrix} \subsetneq \mathbf{\Delta}^0_2(X) \subsetneq \begin{matrix}\mathbf{\Sigma}^0_2(X)\\ \mathbf{\Pi}^0_2(X)\end{matrix} \subsetneq \cdots$$

Proof. Let U be an X-universal set for $\mathbf{\Sigma}^0_\xi(X)$, and $A := \operatorname{AntiDiag}(U)$. Then by definition $A = \delta^{-1}(U^c)$ with $\delta : X \to X \times X$ given by $\delta(x) = (x, x)$. Moreover, $A \in \mathbf{\Pi}^0_\xi(X)$ since $U^c \in \mathbf{\Pi}^0_\xi(X \times X)$, δ is continuous, and $\mathbf{\Pi}^0_\xi$ is closed under continuous preimages (Proposition 1.13(a)). But A is not in $\mathbf{\Sigma}^0_\xi(X)$ because A is not a fiber of U (Diagonalization Lemma), yet U parametrizes $\mathbf{\Sigma}^0_\xi(X)$. □

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2 Change of Topology

2.1 Turning Borel Sets into Clopen Sets

Notation. To be precise, we denote a Polish space X by $(X, \mathcal{T})$, where $\mathcal{T}$ is the topology of X. We have the following most useful tool with Borel sets.

Theorem 2.1

Let $(X, \mathcal{T})$ be a Polish space. For each Borel set $A \subseteq X$, there is a finer Polish topology $\mathcal{T}_A$ with respect to which A is clopen and $\mathcal{B}(\mathcal{T}_A) = \mathcal{B}(\mathcal{T})$.

We prove two lemmas first.

Lemma 2.2

Let $(X, \mathcal{T})$ be a Polish space, $F \subseteq X$ a closed set. Then the topology $\mathcal{T}_F$ generated by $\mathcal{T} \cup \{F\}$ is Polish, with F clopen in $\mathcal{T}_F$ and $\mathcal{B}(\mathcal{T}_F) = \mathcal{B}(\mathcal{T})$.

Proof. We just need to show $\mathcal{T}_F$ is Polish. Observe that $\mathcal{T}_F$ is nothing but the disjoint union topology of F and $X \setminus F$ with their relative topologies respectively, both of which are Polish since F and $X \setminus F$ are both G_δ subsets of the Polish space $(X, \mathcal{T})$, i.e. the map

F	$X \setminus F$

$$(X, \mathcal{T}_F) \to \{0\} \times F \sqcup \{1\} \times (X \setminus F), \qquad x \mapsto \begin{cases}(0, x) & \text{if } x \in F;\\ (1, x) & \text{otherwise}\end{cases}$$

is (trivially) a homeomorphism. But we showed on day 1 (Part 1, Proposition 1.4(b)) that a disjoint union of Polish spaces is Polish (with respect to $d((n,x),(m,y)) := d_n(x,y)$ if $n = m$; 1 otherwise). Hence $(X, \mathcal{T}_F)$

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is Polish. In particular, $\mathcal{B}(X, \mathcal{T}_F) = \mathcal{B}(X, \mathcal{T})$ since F is closed, hence Borel in $\mathcal{T}$. □

Lemma 2.3

Let $(X, \mathcal{T})$ be a Polish space, $(\mathcal{T}_n)_{n\in\mathbb{N}}$ a sequence of Polish topologies on X refining $\mathcal{T}$ with $\mathcal{T}_n \subseteq \mathcal{B}(\mathcal{T})$ (i.e. $\mathcal{B}(\mathcal{T}_n) = \mathcal{B}(\mathcal{T})$). Then the topology $\mathcal{T}_\infty$ generated by $\bigcup_{n\in\mathbb{N}} \mathcal{T}_n$ is Polish with $\mathcal{B}(\mathcal{T}_\infty) = \mathcal{B}(\mathcal{T})$.

Proof. Take a countable basis $\mathcal{U}_n$ for each $\mathcal{T}_n$. Then $\mathcal{T}_\infty$ is generated by the finite intersections of sets from $\bigcup_n \mathcal{U}_n$. So every open set in $\mathcal{T}_\infty$ is some countable union (by second countability of $\mathcal{T}_\infty$) of finite intersections of sets from $\bigcup_n \mathcal{U}_n$, hence $\mathcal{B}(\mathcal{T})$. Thus $\mathcal{T}_\infty \subseteq \mathcal{B}(\mathcal{T}) \implies \mathcal{B}(\mathcal{T}_\infty) \subseteq \mathcal{B}(\mathcal{T})$ since $\mathcal{B}(\mathcal{T}_\infty)$ is the smallest σ-algebra containing $\mathcal{T}_\infty$ (and $\supseteq$ holds since $\mathcal{T} \subseteq \mathcal{T}_\infty$).

It is left to show that $(X, \mathcal{T}_\infty)$ admits a complete compatible metric. We show that $(X, \mathcal{T}_\infty)$ is homeomorphic to the diagonal $\Delta := \{(x)_{n\in\mathbb{N}} : x \in X\} \subseteq X^{\mathbb{N}}$ equipped with the product topology, via

$$\delta : (X, \mathcal{T}_\infty) \to \Big(X^{\mathbb{N}}, \prod_{n\in\mathbb{N}} \mathcal{T}_n\Big), \qquad x \mapsto (x)_n = (x, x, \dots).$$

Clearly δ is a bijection onto Δ. δ is continuous since each coordinate map $\delta_n := \mathrm{proj}_n \circ \delta : (X, \mathcal{T}_\infty) \to (X, \mathcal{T}_n)$ is continuous: δ_n is simply the identity map from a finer topology to a coarser topology, hence continuous. Then it is easy to check that the coordinate maps δ_n being

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continuous $\implies$ δ continuous. On the other hand, to show $\delta^{-1} : \Delta \to (X, \mathcal{T}_\infty)$ is continuous, it suffices to show that preimages of open sets in $\mathcal{T}_n$ by δ^{-1} are open, since $\mathcal{T}_\infty$ is generated by $\bigcup_n \mathcal{T}_n$, and preimages commute with $\cap$ and $\cup$. But for any $n \in \mathbb{N}$, $\delta^{-1} = \mathrm{proj}_n|_\Delta$, hence for $U \in \mathcal{T}_n$, $(\delta^{-1})^{-1}(U) = (\mathrm{proj}_n|_\Delta)^{-1}(U)$ is open in the product topology since the proj_n are continuous by definition of the product topology.

Therefore $(X, \mathcal{T}_\infty) \xrightarrow{\delta} \Delta \subseteq X^{\mathbb{N}}$ is a homeomorphism. Furthermore, Δ is closed since each $\mathcal{T}_n$ is Hausdorff (exercise: show the complement is open). Hence Δ is Polish since it is a G_δ subset of a Polish space (a countable product of Polish spaces is Polish, Part 1, Proposition 1.4(c), and closed sets are G_δ, Part 1, Theorem 1.9). Thus $(X, \mathcal{T}_\infty)$ is Polish. □

Proof of Theorem 2.1 (Clopenification). We do the same σ-algebra trick. Let

$$\mathcal{S} := \{A \in \mathcal{B}(\mathcal{T}) : \exists \text{ Polish } \mathcal{T}_A \supseteq \mathcal{T} \text{ with } A \text{ clopen in } \mathcal{T}_A \text{ and } \mathcal{B}(\mathcal{T}_A) = \mathcal{B}(\mathcal{T})\}.$$

$\mathcal{S} \supseteq$ all closed sets by Lemma 2.2. $\mathcal{S}$ is closed under complements since clopenness is. Finally, for $A_n \in \mathcal{S}$, there are Polish $\mathcal{T}_n \supseteq \mathcal{T}$, $\mathcal{B}(\mathcal{T}_n) = \mathcal{B}(\mathcal{T})$, with A_n clopen in $\mathcal{T}_n$; by Lemma 2.3, $\mathcal{T}_\infty$ is Polish with $\bigcup_n A_n$ open in $\mathcal{T}_\infty$. Lastly apply Lemma 2.2 again (to the closed set $\big(\bigcup_n A_n\big)^c$) so that $\bigcup_n A_n$ is clopen in $\widetilde{\mathcal{T}}_\infty \supseteq \mathcal{T}_\infty$ with $\mathcal{B}(\widetilde{\mathcal{T}}_\infty) = \mathcal{B}(\mathcal{T}_\infty) = \mathcal{B}(\mathcal{T})$. Hence $\bigcup_n A_n \in \mathcal{S}$, i.e. $\mathcal{S}$ is

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closed under countable $\cup$, hence a σ-algebra. But open sets ($= \mathcal{T}$) are in $\mathcal{S}$ (since closed sets are) and $\mathcal{S} \subseteq \mathcal{B}(\mathcal{T}) \implies \langle \mathcal{T} \rangle_\sigma = \mathcal{B}(\mathcal{T}) = \mathcal{S}$. □

Remark 2.4 (Caution)

We proved that a closed (thus the same for an open) set can be made clopen by simply taking the topology $\mathcal{T}_F$ generated by $\mathcal{T} \cup \{F\}$. However, it is not true that if B is a Borel set in a Polish space $(X, \mathcal{T})$, then the topology generated by $\mathcal{T} \cup \{B, B^c\}$ is Polish. To obtain a Polish topology in which B is clopen, one has to clopenify all (countably many) sets that go into the definition of B from open / closed sets (what we did in Lemma 2.3). □

Corollary 2.5

Let $\mathcal{U}$ be a countable collection of Borel sets in a Polish space $(X, \mathcal{T})$. Then there is a Polish topology $\mathcal{T}' \supseteq \mathcal{T}$ in which every set in $\mathcal{U}$ is clopen and $\mathcal{B}(\mathcal{T}') = \mathcal{B}(\mathcal{T})$.

Proof. For each $U \in \mathcal{U}$, let $\mathcal{T}_U \supseteq \mathcal{T}$ be a Polish topology in which U is clopen. Then by Lemma 2.3 above, the topology generated by $\bigcup_{U \in \mathcal{U}} \mathcal{T}_U$ is Polish with the desired property. □

Definition 2.6

Recall that a topological space X has the perfect set property (PSP) if X is either countable or there is a nonempty perfect subset (i.e. perfect closed) of X.

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If X is Polish, then X has the PSP iff X is countable or $2^{\mathbb{N}}$ topologically embeds into X: ($\Rightarrow$) is by Cantor's perfect set theorem from Part 1, §4. ($\Leftarrow$) since $2^{\mathbb{N}}$ is compact, thus compact in the Hausdorff space X, hence closed in X (Part 1, Theorem 3.7(b)). Thus $2^{\mathbb{N}}$ is a nonempty perfect (closed) subset of X. □

We now give the corollary that not only Polish spaces and Polish subspaces (i.e. G_δ subsets of Polish spaces) have the PSP, but the Borel sets also have it.

Corollary 2.7

Every Borel set B in a Polish space $(X, \mathcal{T})$ has the perfect set property.

Proof. Take a Polish topology $\mathcal{T}' \supseteq \mathcal{T}$ on X in which B is clopen, hence Polish in the subspace topology of $\mathcal{T}'$. By the Cantor–Bendixson theorem, B has the PSP, so if B is uncountable, then there is a continuous embedding $f : 2^{\mathbb{N}} \hookrightarrow (B, \mathcal{T}'|_B)$. Then trivially, f is still continuous as a map to $(B, \mathcal{T}|_B)$ and still injective, hence an embedding into $(B, \mathcal{T}|_B)$ because $2^{\mathbb{N}}$ is compact and $\mathcal{T}|_B$ is Hausdorff (Part 1, Theorem 3.7(e)). □

Corollary 2.8 (Making Borel functions continuous)

Let $(X, \mathcal{T})$ be a Polish space and Y be a second countable topological space. Let $f : X \to Y$ be a Borel (measurable) function. Then there is a Polish topology $\mathcal{T}_f \supseteq \mathcal{T}$ such that $f : (X, \mathcal{T}_f) \to Y$ is continuous and $\mathcal{B}(\mathcal{T}_f) = \mathcal{B}(\mathcal{T})$.

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Proof. Let $\mathcal{U}$ be a countable basis for Y, and $\mathcal{T}_f \supseteq \mathcal{T}$ a Polish topology with $\mathcal{B}(\mathcal{T}_f) = \mathcal{B}(\mathcal{T})$ in which every set $f^{-1}(U)$ is clopen, for all $U \in \mathcal{U}$. Hence f is continuous from $(X, \mathcal{T}_f)$ to Y. □

Corollary 2.9

Let $(X, \mathcal{T})$ be a Polish space, $B \subseteq X$ a Borel subset. Then there is a continuous bijection $f : F \to B$ for some closed $F \subseteq \mathbb{N}^{\mathbb{N}}$. In particular, if $B \neq \varnothing$, then there is a continuous surjection $\mathbb{N}^{\mathbb{N}} \twoheadrightarrow B$.

Proof. Let $\mathcal{T}' \supseteq \mathcal{T}$ be a Polish topology with B clopen. Hence $(B, \mathcal{T}'|_B)$ is Polish, so there is a continuous bijection $f : F \xrightarrow{\sim} (B, \mathcal{T}'|_B)$, where $F \subseteq \mathbb{N}^{\mathbb{N}}$ is closed. If $B \neq \varnothing$, then there is a continuous surjection $g : \mathbb{N}^{\mathbb{N}} \twoheadrightarrow (B, \mathcal{T}'|_B)$, composing f with a retraction $\mathbb{N}^{\mathbb{N}} \twoheadrightarrow F$. But then f, g are trivially continuous with respect to the coarser topology $\mathcal{T}|_B$. □

Proposition 2.10

Let $(X, \mathcal{T})$ be a Polish space. Then there exists a zero-dimensional Polish topology $\mathcal{T}_0 \supseteq \mathcal{T}$ such that $\mathcal{B}(\mathcal{T}_0) = \mathcal{B}(\mathcal{T})$.

Proof. HW 3. □

Proposition 2.11

Let Γ be a countable group, and consider a Borel action of Γ on a Polish space $(X, \mathcal{T})$, i.e. every $g \in \Gamma$ acts as a Borel automorphism of X. Then there exists a Polish topology $\mathcal{T}_0 \supseteq \mathcal{T}$ with $\mathcal{B}(\mathcal{T}_0) = \mathcal{B}(\mathcal{T})$ that makes the action of Γ continuous. Moreover, $\mathcal{T}_0$ can be taken to be zero-dimensional.

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Proof. HW 3. □

2.2 Making Closed-Graph Functions Continuous for Free

Definition 2.12

Let $(X, \mathcal{T}_X)$, $(Y, \mathcal{T}_Y)$ be topological spaces, $f : X \to Y$. The topology generated by f over $\mathcal{T}_X$ is the coarsest topology $\mathcal{T}_{X,f} \supseteq \mathcal{T}_X$ which makes f continuous. □

Observation 2.13

$\mathcal{T}_{X,f}$ is the topology generated by $\mathcal{T}_X \cup \{f^{-1}(V) : V \in \mathcal{T}_Y\}$. Every $U \in \mathcal{T}_{X,f}$ is an arbitrary union of sets of the form $U \cap f^{-1}(V)$, $U \in \mathcal{T}_X$, $V \in \mathcal{T}_Y$. □

Lemma 2.14

Let $(X, \mathcal{T}_X)$, $(Y, \mathcal{T}_Y)$ be topological spaces and $f : X \to Y$. Then we have two topologies on $X \times Y$: $\mathcal{T}_X \times \mathcal{T}_Y$ and $\mathcal{T}_{X,f} \times \mathcal{T}_Y$. On the graph $G := \operatorname{graph}(f) = \{(x, f(x)) : x \in X\} \subseteq X \times Y$, the relative topologies with respect to $\mathcal{T}_X \times \mathcal{T}_Y$ and $\mathcal{T}_{X,f} \times \mathcal{T}_Y$ coincide.

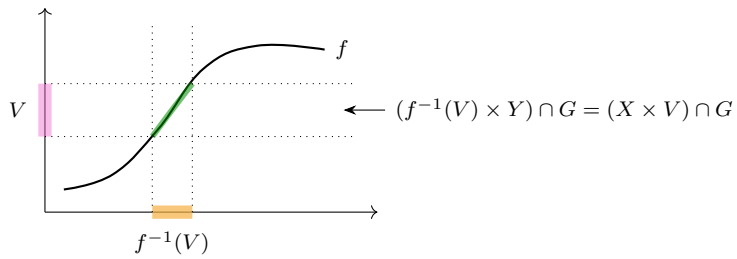
Proof. Notice that the basic open sets in $\mathcal{T}_{X,f} \times \mathcal{T}_Y|_G$ have the form

$$[(U \cap f^{-1}(V)) \times W] \cap G = [(U \times W) \cap G] \cap [(f^{-1}(V) \times W) \cap G].$$

But $U \times W$ is open in $\mathcal{T}_X \times \mathcal{T}_Y$, thus $(U \times W) \cap G$ is open in $\mathcal{T}_X \times \mathcal{T}_Y|_G$. It is left to show that $(f^{-1}(V) \times W) \cap G$ is open in $\mathcal{T}_X \times \mathcal{T}_Y|_G$. But this is simply because

$$(f^{-1}(V) \times W) \cap G = [X \times (V \cap W)] \cap G,$$

which is open in $\mathcal{T}_X \times \mathcal{T}_Y|_G$ since $X \in \mathcal{T}_X$ and $V \cap W \in \mathcal{T}_Y$.



□

Lemma 2.15 (Continuity via graph)

Let X, Y be topological spaces. A

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function $f : X \to Y$ is continuous

$$\iff \pi_f : X \to \operatorname{graph}(f),\ x \mapsto (x, f(x)) \text{ is a homeomorphism} \iff \pi_f \text{ is continuous.}$$

Proof. If π_f is continuous, then so is f since $f = \mathrm{proj}_Y \circ \pi_f$. On the other hand, if f is continuous, then so is π_f since each coordinate function $x \mapsto x$, $x \mapsto f(x)$ is continuous, hence continuity of π_f by definition of the product topology. Lastly, we get π_f^{-1} continuous for free since $\pi_f^{-1} = \mathrm{proj}_X|_{\mathrm{graph}(f)}$. □

Proposition 2.16 (Ruiyuan Chen)

Let $(X, \mathcal{T}_X)$, $(Y, \mathcal{T}_Y)$ be Polish spaces, $f : X \to Y$. If $\mathrm{graph}(f)$ is closed in $\mathcal{T}_X \times \mathcal{T}_Y$ (for instance, if f is continuous with respect to some coarser Hausdorff topology on Y), then $\mathcal{T}_{X,f}$ is Polish.

Proof. $f : (X, \mathcal{T}_{X,f}) \to Y$ is continuous by definition of $\mathcal{T}_{X,f}$. Set $G := \mathrm{graph}(f)$. Then $\pi_f : (X, \mathcal{T}_{X,f}) \to (G, \mathcal{T}_{X,f} \times \mathcal{T}_Y|_G)$ is a homeomorphism. Then $(G, \mathcal{T}_X \times \mathcal{T}_Y|_G)$ is Polish, being a closed subset of the Polish space $(X \times Y, \mathcal{T}_X \times \mathcal{T}_Y)$, and $\mathcal{T}_X \times \mathcal{T}_Y|_G = \mathcal{T}_{X,f} \times \mathcal{T}_Y|_G$ by Lemma 2.14. Thus $(X, \mathcal{T}_{X,f}) \cong (G, \mathcal{T}_X \times \mathcal{T}_Y|_G) = (G, \mathcal{T}_{X,f} \times \mathcal{T}_Y|_G)$ is Polish. □

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3 Analytic Sets

3.1 Definition and Closure Properties

We know that the class of Borel sets is closed under continuous / Borel preimages. Lebesgue had a false proof that it is closed under continuous images. It was discovered by Souslin years later, who proved that these are new sets and studied them with his PhD advisor Luzin.

Definition 3.1

A subset A of a Polish space X is called analytic if it is a continuous image of a Borel subset of some Polish space, i.e. if there are a Polish space Y, a Borel subset $B \subseteq Y$, and a continuous function $f : Y \to X$ such that $A = f(B)$. □

Proposition 3.2 (Equivalence of analyticity)

Let X be Polish, $\varnothing \neq A \subseteq X$. Then the following are equivalent:

(1) A is analytic (the definition above).
(2) There are a Polish space Y and a continuous $f : Y \to X$ such that $f(Y) = A$.
(3) There is a continuous $f : \mathbb{N}^{\mathbb{N}} \to X$ such that $f(\mathbb{N}^{\mathbb{N}}) = A$.
⋆(4) There is a closed $F \subseteq X \times \mathbb{N}^{\mathbb{N}}$ such that $\mathrm{proj}_X(F) = A$.
(5) There are a Polish Y and a Borel $B \subseteq X \times Y$ such that $\mathrm{proj}_X(B) = A$.
(6) There are a Polish Y, a Borel $B \subseteq Y$ and a Borel (measurable) $f : Y \to X$ such that $f(B) = A$.

Proof. Immediately, (4) ⇒ (5) ⇒ (1) and (1) ⇒ (6). (1) ⇒ (2) follows by a change of topology to make B clopen, hence Polish with a finer topology $\mathcal{T}_B$, but then f is just more continuous. (2) ⇒ (3)

since there is a continuous surjection $\mathbb{N}^{\mathbb{N}} \twoheadrightarrow Y$, thus we can

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compose it with f. (3) $\Rightarrow$ (4) follows by taking

$$F = \operatorname{graph}(f)^{-1} := \{(f(x), x) : x \in \mathbb{N}^{\mathbb{N}}\}$$

since the graph of a continuous function (into a Hausdorff space) is closed, and $A = f(\mathbb{N}^{\mathbb{N}}) = \operatorname{proj}_X(F)$. (6) $\Rightarrow$ (1) since we can make a Borel measurable function into a continuous function by Corollary 2.8. □

Notation. We denote by $\mathbf{\Sigma}^1_1$ the class of all analytic subsets of Polish spaces, so given a Polish space X, the set of analytic subsets of X is denoted by $\mathbf{\Sigma}^1_1(X)$. Its dual class is denoted by $\mathbf{\Pi}^1_1 := \neg\mathbf{\Sigma}^1_1$, with

$$\mathbf{\Pi}^1_1(X) = \neg\mathbf{\Sigma}^1_1(X) := \{A^c : A \in \mathbf{\Sigma}^1_1(X)\},$$

its elements called coanalytic sets. We also have the ambiguous class $\mathbf{\Delta}^1_1 := \mathbf{\Pi}^1_1 \cap \mathbf{\Sigma}^1_1$.

By (4) above, we have that $\mathbf{\Sigma}^1_1 = \exists^{\mathbb{N}^{\mathbb{N}}}\mathbf{\Pi}^0_1$, i.e. $\mathbf{\Sigma}^1_1(X) = \exists^{\mathbb{N}^{\mathbb{N}}}\mathbf{\Pi}^0_1(X)$, as we discussed in Remark 1.16. Taking its dual, we get $\mathbf{\Pi}^1_1(X) = \forall^{\mathbb{N}^{\mathbb{N}}}\mathbf{\Sigma}^0_1(X)$. This is what will allow us to build a universal set for $\mathbf{\Sigma}^1_1$, hence an analytic set which is not Borel, via Cantor's diagonalization. □

Observation 3.3 (Borel sets are bi-analytic)

$\mathcal{B}(X) \subseteq \mathbf{\Delta}^1_1(X)$ since by (6), taking $f = \mathrm{id}$, all Borel sets are analytic; but $\mathcal{B}(X)$ is closed under taking complements, hence they are also coanalytic. □

Proposition 3.4

The class $\mathbf{\Sigma}^1_1$ is closed under

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(i) continuous images and preimages (in fact, Borel images and preimages);
(ii) countable unions and countable intersections.

Proof. (i) Let X, Y be Polish spaces, $g : X \to Y$ continuous, $\varnothing \neq A \subseteq X$ an analytic subset. Then there is a continuous $f : \mathbb{N}^{\mathbb{N}} \to X$ with $f(\mathbb{N}^{\mathbb{N}}) = A$. Thus $g(A) \subseteq Y$ is witnessed by $g \circ f : \mathbb{N}^{\mathbb{N}} \to Y$.

Now suppose $B \subseteq Y$ is analytic. Then there is a closed $F \subseteq Y \times \mathbb{N}^{\mathbb{N}}$ such that $B = \operatorname{proj}_Y(F)$. Define $\tilde{g} : X \times \mathbb{N}^{\mathbb{N}} \to Y \times \mathbb{N}^{\mathbb{N}}$ as $\tilde{g}((x, n)) := (g(x), n)$, which is continuous since its component maps are. Hence $\tilde{g}^{-1}(F)$ is closed. Moreover, observe that

$$\begin{aligned} x \in g^{-1}(B) &\iff g(x) \in B = \operatorname{proj}_Y(F) \iff \exists\, n \in \mathbb{N}^{\mathbb{N}} \text{ such that } (g(x), n) \in F \\ &\iff \exists\, n \in \mathbb{N}^{\mathbb{N}} \text{ such that } (x, n) \in \tilde{g}^{-1}(F) \iff x \in \operatorname{proj}_X\left(\tilde{g}^{-1}(F)\right). \end{aligned}$$

Thus $g^{-1}(B) = \operatorname{proj}_X\left(\tilde{g}^{-1}(F)\right)$, with the latter an analytic set by condition (4) of Proposition 3.2 and the fact that $\tilde{g}^{-1}(F)$ is closed (since $\tilde{g}$ is continuous). Borel functions follow from the fact that they can be made continuous in a finer Polish topology on X.

(ii) Countable intersections: Let X be a Polish space, $A_n \subseteq X$ analytic. Then there are closed $C_n \subseteq X \times \mathbb{N}^{\mathbb{N}}$ such that $A_n = \operatorname{proj}_X(C_n)$, i.e. for each $x \in X$, $x \in A_n \iff \exists y \in \mathbb{N}^{\mathbb{N}}$, $(x,y) \in C_n$. Let $A := \bigcap_{n\in\mathbb{N}} A_n$. Then for $x \in X$,

$$x \in A \iff \forall n \in \mathbb{N},\ \exists y_n \in \mathbb{N}^{\mathbb{N}},\ (x,y_n) \in C_n \iff \exists y \in (\mathbb{N}^{\mathbb{N}})^{\mathbb{N}}\ \forall n \in \mathbb{N},\ (x,y(n)) \in C_n.$$

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Hence with $\widetilde{C}_n := \{(x,y) \in X \times (\mathbb{N}^{\mathbb{N}})^{\mathbb{N}} : (x,y(n)) \in C_n\}$, we get the following:

(a) $A = \operatorname{proj}_X\left(\bigcap_n \widetilde{C}_n\right)$ (trivial).

(b) The $\widetilde{C}_n$ are closed: Observe that the map $\pi_n : X \times (\mathbb{N}^{\mathbb{N}})^{\mathbb{N}} \to X \times \mathbb{N}^{\mathbb{N}}$, $(x,y) \mapsto (x,y(n))$, is continuous since each component map is a projection, and $\widetilde{C}_n = \pi_n^{-1}(C_n)$ with C_n closed, hence the $\widetilde{C}_n$ are closed.

Thus $\bigcap_n \widetilde{C}_n$ is closed, thus A is analytic by (a) and the fact that $\mathbb{N}^{\mathbb{N}} \cong \mathbb{N}^{\mathbb{N}\times\mathbb{N}} \cong (\mathbb{N}^{\mathbb{N}})^{\mathbb{N}}$. We leave the countable union as an exercise (HW 3). □

3.2 Universal Sets for Analytic / Coanalytic Sets

In the same way we proved the strictness of the Borel hierarchy in uncountable Polish spaces with universal sets and Cantor's diagonalization, we show the analog for analytic / coanalytic sets, that $\mathbf{\Pi}^1_1 \neq \mathbf{\Sigma}^1_1$.

Theorem 3.5 (Souslin)

For every uncountable Polish Y and an arbitrary Polish X, there is a Y-universal set for $\mathbf{\Sigma}^1_1(X)$. The same holds for $\mathbf{\Pi}^1_1(X)$.

Proof. We use $\mathbf{\Sigma}^1_1 = \exists^{\mathbb{N}^{\mathbb{N}}}\mathbf{\Pi}^0_1$. Let $F \subseteq Y \times (X \times \mathbb{N}^{\mathbb{N}})$ be a Y-universal set for $\mathbf{\Pi}^0_1(X \times \mathbb{N}^{\mathbb{N}})$. Let $U := \operatorname{proj}_{Y\times X}(F)$. Then we can verify that U is Y-universal for $\mathbf{\Sigma}^1_1(X)$

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because fibers commute with projections:

$$\{U_y : y \in Y\} = \{\operatorname{proj}_X(F_y) : y \in Y\} = \{\operatorname{proj}_X(C) : C \in \mathbf{\Pi}^0_1(X \times \mathbb{N}^{\mathbb{N}})\} = \mathbf{\Sigma}^1_1(X)$$

by definition, and $U = \operatorname{proj}_{Y\times X}(F) \in \mathbf{\Sigma}^1_1(Y \times X)$ by definition as well. □

Corollary 3.6 (Souslin)

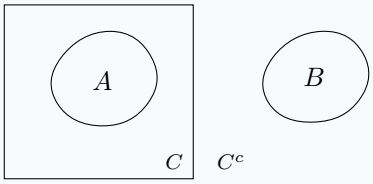
For each uncountable Polish space X, $\mathbf{\Sigma}^1_1(X) \neq \mathbf{\Pi}^1_1(X)$. In particular, $\mathcal{B}(X) \subseteq \mathbf{\Delta}^1_1(X) \subsetneq \mathbf{\Sigma}^1_1(X), \mathbf{\Pi}^1_1(X)$.

Proof. Let $U \subseteq X \times X$ be an X-universal set for $\mathbf{\Sigma}^1_1(X)$. Then $A := \operatorname{AntiDiag}(U) = \{x \in X : (x,x) \notin U\}$ is coanalytic because $A = \delta^{-1}(U^c) = \delta^{-1}(U)^c$, where $\delta : X \to X^2$, $x \mapsto (x,x)$, is continuous and $\mathbf{\Sigma}^1_1$ is closed under continuous preimages. But A is not analytic because $A \neq U_x$, $\forall x \in X$ (Diagonalization Lemma), while $\{U_x : x \in X\} = \mathbf{\Sigma}^1_1(X)$. □

3.3 Analytic Separation

Theorem 3.7 (Luzin)

Let X be Polish and $A, B \subseteq X$ be disjoint analytic sets. Then A, B are Borel-separable, i.e. there is a Borel $C \subseteq X$ such that $C \supseteq A$ and $C^c \supseteq B$.

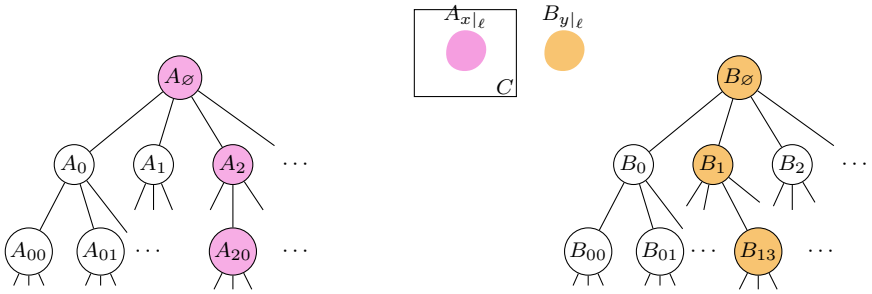


Proof. For a set $P \subseteq X$, observe that the sets which are Borel-separable from P form a σ-ideal, i.e. if each Q_n is Borel-separable from P by a Borel set B_n, then P is Borel-separable from $\bigcup_n Q_n$ by $\bigcup_n B_n$: $\bigcup_n B_n \supseteq \bigcup_n Q_n$ and $(\bigcup_n B_n) \cap P = \bigcup_n (B_n \cap P) = \varnothing$. Hence moreover, if

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P_n, Q_m are Borel-separable for all $m, n \in \mathbb{N}$, then $\bigcup_n P_n$ and $\bigcup_m Q_m$ are Borel-separable: separate P_n from $\bigcup_m Q_m$ for each n, then separate $\bigcup_n P_n$ from $\bigcup_m Q_m$.

Now fix nonempty disjoint analytic sets A, B (if one is empty, then the case is trivial). Then there are continuous surjections $f : \mathbb{N}^{\mathbb{N}} \twoheadrightarrow A$ and $g : \mathbb{N}^{\mathbb{N}} \twoheadrightarrow B$. For $s \in \mathbb{N}^{<\mathbb{N}}$, put $A_s := f([s]_{\mathbb{N}})$, $B_s := g([s]_{\mathbb{N}})$, so $A_s = \bigcup_{n \in \mathbb{N}} A_{sn}$ and $B_s = \bigcup_{n \in \mathbb{N}} B_{sn}$, and $A = A_\varnothing$ and $B = B_\varnothing$. If A, B are not separable, then since $A = A_\varnothing = \bigcup_n A_n$, $B = B_\varnothing = \bigcup_m B_m$, there are $n_0, m_0 \in \mathbb{N}$ such that A_{n_0} and B_{m_0} are not Borel-separable. Further, since $A_s = \bigcup_n A_{sn}$, $B_s = \bigcup_n B_{sn}$, there are $n_1, m_1 \in \mathbb{N}$ such that $A_{n_0 n_1}$ and $B_{m_0 m_1}$ are not Borel-separable. Hence by induction, there are infinite branches $x, y \in \mathbb{N}^{\mathbb{N}}$ such that for every $\ell \in \mathbb{N}$, $A_{x|_\ell}$, $B_{y|_\ell}$ are not Borel-separable.



But $f(x)$ and $g(y)$ are distinct because $f(x) \in A$ and $g(y) \in B$, hence separable by open sets by Hausdorffness, i.e. there are $U, V \subseteq X$ disjoint open such that $U \ni f(x)$ and $V \ni g(y)$. By

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the continuity of f and g and the fact that $([x|_\ell])_\ell$ and $([y|_\ell])_\ell$ form neighborhood bases at x, y respectively, there is $\ell \in \mathbb{N}$ such that $[x|_\ell] \subseteq f^{-1}(U)$, $[y|_\ell] \subseteq g^{-1}(V)$. But then $A_{x|_\ell} = f([x|_\ell])$ and $B_{y|_\ell} = g([y|_\ell])$ are Borel-separable by either U or V. $\Rightarrow\Leftarrow$ □

Corollary 3.8 (Souslin)

Let X be Polish. Then $\mathcal{B}(X) = \mathbf{\Delta}^1_1(X)$.

Proof. It is left to prove $\mathbf{\Delta}^1_1(X) \subseteq \mathcal{B}(X)$ (the other inclusion is the observation above). Take $A \in \mathbf{\Delta}^1_1(X)$. Then both A and A^c are analytic. Hence by Luzin's analytic separation, there is $B \in \mathcal{B}(X)$ such that $B \supseteq A$, $B^c \supseteq A^c$. Thus $B = A$ and $A \in \mathcal{B}(X)$. □

Remark. This gives a powerful method for proving that a set is Borel, i.e. we just check analyticity and coanalyticity. □

Corollary 3.9

Let X, Y be Polish, $f : X \to Y$. The following are equivalent:

(1) f is Borel, i.e. f-preimages of Borel (equivalently open) sets are Borel.
(2) $\operatorname{graph}(f) \in \mathcal{B}(X \times Y)$, i.e. the graph is Borel.
(3) $\operatorname{graph}(f) \in \mathbf{\Sigma}^1_1(X \times Y)$, i.e. the graph is analytic.

Proof. (1) ⇒ (2) Fix a countable basis $\{V_n\}_{n\in\mathbb{N}}$ for Y and note that $\forall (x, y) \in X \times Y$,

$$\begin{aligned}(x, y) \in \operatorname{graph}(f) &\iff f(x) = y \iff \forall n \in \mathbb{N}\ (y \in V_n \implies x \in f^{-1}(V_n)) \\ &\iff \underbrace{\forall n \in \mathbb{N}}_{\text{countable}\ \cap}\ \big(\ \underbrace{y \notin V_n}_{\text{closed } (V_n^c)}\ \text{ or }\ \underbrace{x \in f^{-1}(V_n)}_{\text{Borel}}\big).\end{aligned}$$

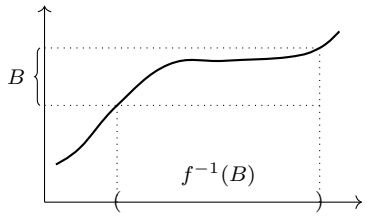
(2) ⇒ (3) is trivial.

(3) ⇒ (1) Assume $\operatorname{graph}(f)$ is analytic and fix a Borel $B \subseteq$

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Y to show that $f^{-1}(B)$ is Borel. For each $x \in X$, we have

$$\begin{aligned}x \in f^{-1}(B) &\iff \exists y \in Y\ \big(f(x) = y \text{ and } y \in B\big) \\ &\iff \exists y \in Y\ \big[(x, y) \in \operatorname{graph}(f)\ \wedge\ (x, y) \in X \times B\big].\end{aligned}$$



Thus

$$f^{-1}(B) = \underbrace{\operatorname{proj}_X}_{\text{continuous}} \Big[\underbrace{\operatorname{graph}(f)}_{\mathbf{\Sigma}^1_1} \cap \underbrace{(X \times B)}_{= \underbrace{\operatorname{proj}_Y^{-1}}_{\text{continuous}} \underbrace{(B)}_{\text{Borel}}} \Big]$$

$\mathbf{\Sigma}^1_1$ by closure under (countable) finite intersections

Borel, since $\mathcal{B}$ is closed under continuous preimages

is analytic, since $\mathbf{\Sigma}^1_1$ is closed under continuous images. But then $f^{-1}(B)^c = f^{-1}(B^c)$ is analytic, so $f^{-1}(B) \in \mathbf{\Delta}^1_1(X) = \mathcal{B}(X)$ is Borel by Souslin's theorem. □

Corollary 3.10

Let X be Polish, $\{A_n\}_{n\in\mathbb{N}}$ a family of pairwise disjoint analytic subsets of X. Then there are pairwise disjoint Borel sets $\{B_n\}_{n\in\mathbb{N}}$ such that $B_n \supseteq A_n$, $\forall n \in \mathbb{N}$.

Proof. Exercise. Hint: Use finite intersections to construct Borel separations for each A_n, then disjointify them. □

3.4 Souslin's Operation $\mathcal{A}$

We now introduce a powerful machinery that equivalently defines the analytic sets. It further allows us to prove that all analytic sets are universally measurable, i.e. measurable with respect to an arbitrary complete σ-finite

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measure μ that measures all Borel sets, i.e. $\mathcal{M} \supseteq \mathcal{B}(X)$, where $\mathcal{M}$ denotes the μ-measurable sets.

Definition 3.11

For a set X and a pruned tree $T \subseteq \mathbb{N}^{<\mathbb{N}}$, we call any sequence $(P_s)_{s\in T}$ of subsets of X a Souslin scheme on X. Call it monotone if $P_t \subseteq P_s$ for all $t \supseteq s$. Call it proper if it is monotone and $P_{si} \cap P_{sj} = \varnothing$, $\forall\, si, sj \in T$, $i \neq j \in \mathbb{N}$. □

Definition 3.12

Souslin's operation $\mathcal{A}$ applied to a Souslin scheme $(P_s)_{s\in T}$ on X results in the set

$$\mathcal{A}(P_s)_{s\in T} := \bigcup_{y\in[T]} \bigcap_{n\in\mathbb{N}} P_{y|_n}. \quad \square$$

Observation 3.13

Given a Souslin scheme $(P_s)_{s\in T}$, we can extend T to $\mathbb{N}^{<\mathbb{N}}$ by just setting $P_s := \varnothing$ for $s \in \mathbb{N}^{<\mathbb{N}} \setminus T$. Note that we are not assuming anything about the scheme nor about $\mathcal{A}(P_s)_{s\in T}$. Hence, we might take WLOG our $T = \mathbb{N}^{<\mathbb{N}}$. □

Proposition 3.14

If $(P_s)_{s\in T}$ is a proper Souslin scheme, then

$$\mathcal{A}(P_s)_{s\in T} = \bigcap_{n\in\mathbb{N}} \bigcup_{\substack{s\in T\\ |s|=n}} P_s.$$

Proof. $\subseteq$: If $x \in \mathcal{A}(P_s)_{s\in T}$, there is $y \in [T]$ such that $\forall n$, $x \in P_{y|_n}$. Then $\forall n$, $x \in \bigcup_{s\in T,|s|=n} P_s$.

$\supseteq$: If $x \in \bigcap_{n\in\mathbb{N}} \bigcup_{s\in T,|s|=n} P_s$, then $\forall n$, $\exists!\, s_n \in T$ with $|s_n| = n$ such that $x \in P_{s_n}$, by properness. By monotonicity, $t := s_{n+1}|_n \subseteq s_{n+1}$,

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hence $P_t \supseteq P_{s_{n+1}} \ni x$, thus by uniqueness, $t = s_n$, and $s_n \subseteq s_{n+1}$. Thus defining $y := \bigcup_{n\in\mathbb{N}} s_n \in [T]$ we get $x \in \bigcap_{n\in\mathbb{N}} P_{y|_n} \subseteq \mathcal{A}(P_s)_{s\in T}$. □

Lemma 3.15

If a Souslin scheme $(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$ satisfies $P_s = \bigcup_{i\in\mathbb{N}} P_{si}$, then $\mathcal{A}(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}} = P_\varnothing$.

Proof. By definition, $\mathcal{A}(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}} \subseteq P_\varnothing$. Let $x \in P_\varnothing$; note that there is $y_0 \in \mathbb{N}$ such that $x \in P_{y_0}$, then there is $y_1 \in \mathbb{N}$ such that $x \in P_{y_0 y_1}$, ... By induction we get $y \in \mathbb{N}^\mathbb{N}$ such that $x \in P_{y|_n}$, $\forall n \in \mathbb{N}$, so $x \in \mathcal{A}(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$. □

Definition 3.16

Let Γ be a class of sets in metrizable topological spaces. For each such space X, denote by $\mathcal{A}\Gamma(X)$ the collection of sets of the form $\mathcal{A}(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$ where $P_s \in \Gamma(X)$. The following shows that the operation $\mathcal{A}$ can be viewed as a projection. □

Observation 3.17 (Operation $\mathcal{A}$ as a projection)

For a Souslin scheme $(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$ on a set X, $\mathcal{A}(P_s)_s = \mathrm{proj}_X(P)$, where the set $P \subseteq X \times \mathbb{N}^\mathbb{N}$ is defined by: for each $(x, y) \in X \times \mathbb{N}^\mathbb{N}$,

$$\begin{aligned}(x,y) \in P \;:\iff\; & \forall n \in \mathbb{N},\ x \in P_{y|_n}\\ \iff\; & \forall n \in \mathbb{N}\ \exists s \in \mathbb{N}^{<\mathbb{N}}\ (y|_n = s \ \wedge\ x \in P_s)\\ \iff\; & \forall n \in \mathbb{N}\ \forall s \in \mathbb{N}^{<\mathbb{N}}\ (y|_n = s \implies x \in P_s).\end{aligned}$$

Moreover, if Γ is a class of sets containing all clopen sets and closed under finite unions and countable intersections (e.g. $\Gamma = \mathbf{\Pi}^0_1$ or $\mathbf{\Sigma}^1_1$), then $\mathcal{A}\Gamma \subseteq \exists^{\mathbb{N}^\mathbb{N}}\Gamma$ because

$$(x,y) \in P \iff \underbrace{\forall n \in \mathbb{N}}_{\text{countable}\,\cap}\ \underbrace{\forall s \in \mathbb{N}^{<\mathbb{N}}}_{\text{countable}\,\cap}\ \underbrace{\big(\underbrace{y|_n \neq s}_{\text{clopen}} \vee \underbrace{x \in P_s}_{\Gamma}\big)}_{\text{finite}\,\cup\,\in\,\Gamma}$$

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and $\mathcal{A}(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}} = \mathrm{proj}_X(P)$. In particular, $\mathcal{A}\mathbf{\Sigma}^1_1 \subseteq \exists^{\mathbb{N}^{\mathbb{N}}}\mathbf{\Sigma}^1_1 = \mathbf{\Sigma}^1_1$, with the last equality because $\mathbf{\Sigma}^1_1$ is closed under continuous images (proj) and continuous preimages (proj^{-1}). □

We now show $\mathbf{\Sigma}^1_1 \subseteq \mathcal{A}\mathbf{\Sigma}^1_1$ in Polish spaces.

Lemma 3.18

Let X be Polish and $f : \mathbb{N}^{\mathbb{N}} \to X$ be continuous. Then for each $s \in \mathbb{N}^{<\mathbb{N}}$, $P_s := f([s])$ is analytic and

$$f(\mathbb{N}^{\mathbb{N}}) = \mathcal{A}(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}} = \mathcal{A}(\overline{P_s})_{s\in\mathbb{N}^{<\mathbb{N}}}.$$

In fact, $(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$ satisfies, for all $s \in \mathbb{N}^{<\mathbb{N}}$:

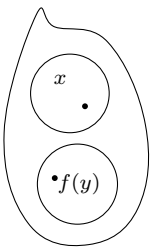
(i) $P_s \neq \varnothing$;
(ii) $P_s = \bigcup_{i\in\mathbb{N}} P_{si}$ (so that Lemma 3.15 applies);
(iii) for each $y \in \mathbb{N}^{\mathbb{N}}$ and open $U \subseteq X$, if $P_y := \bigcap_{n\in\mathbb{N}} P_{y|_n} \subseteq U$, then there is $n \in \mathbb{N}$ such that $P_{y|_n} \subseteq U$;
(iv) the scheme $(\overline{P_s})_{s\in\mathbb{N}^{<\mathbb{N}}}$ has vanishing diameter with respect to any compatible metric on X.

Proof. We leave the verification of the analyticity of P_s and of (i) and (ii) as exercises. (ii) + Lemma 3.15 $\implies$ $\mathcal{A}(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}} = P_\varnothing = f(\mathbb{N}^{\mathbb{N}})$. (iii) is by continuity of f: first notice that $f(y) \in f([y|_n]) = P_{y|_n}$, thus $f(y) \in P_y \subseteq U$. Hence $y \in f^{-1}(U)$, which is open, thus there is $n \in \mathbb{N}$ such that $[y|_n] \subseteq f^{-1}(U)$, i.e. $P_{y|_n} := f([y|_n]) \subseteq U$. (iv) is also by continuity since for $\varepsilon > 0$, there is $n \in \mathbb{N}$ such that $f([y|_n]) \subseteq B_\varepsilon(f(y))$, hence

$$\operatorname{diam}\overline{P_{y|_n}} = \operatorname{diam} P_{y|_n} = \operatorname{diam} f([y|_n]) \leq \operatorname{diam}\left(B_\varepsilon(f(y))\right) \leq 2\varepsilon \to 0.$$

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It is also clear from the definition that $\mathcal{A}(\overline{P_s})_{s\in\mathbb{N}^{<\mathbb{N}}} \supseteq \mathcal{A}(P_s)_s$, hence it remains to show the reverse inclusion. Let $x \in \mathcal{A}(\overline{P_s})_{s\in\mathbb{N}^{<\mathbb{N}}}$, so $x \in \bigcap_{n\in\mathbb{N}} \overline{P_{y|_n}}$ for some $y \in \mathbb{N}^{\mathbb{N}}$. It suffices to show $x = f(y)$, and then $f(y) \in f(\mathbb{N}^{\mathbb{N}}) = P_\varnothing = \mathcal{A}(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$. Suppose otherwise; then by Hausdorffness of X, we have disjoint open $U \ni x$, $V \ni f(y)$, thus $x \notin \overline{V}$. By continuity of f, there is $n \in \mathbb{N}$ such that $f([y|_n]) \subseteq V$, hence $x \notin \overline{f([y|_n])} = \overline{P_{y|_n}} \subseteq \overline{V}$, a contradiction.



□

Theorem 3.19 (Characterization of Σ^1_1 via $\mathcal{A}$)

Let X be Polish and $\varnothing \neq A \subseteq X$. The following are equivalent:

(1) A is analytic ($\iff$ there is a continuous $f : \mathbb{N}^{\mathbb{N}} \to X$ such that $f(\mathbb{N}^{\mathbb{N}}) = A$).
(2) $A = \mathcal{A}(F_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$, each $F_s \neq \varnothing$ closed and the scheme $(F_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$ monotone with vanishing diameter.
(3) $A = \mathcal{A}(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$, each P_s analytic.

As a result,

$$\mathbf{\Sigma}^1_1 = \mathcal{A}\mathbf{\Sigma}^1_1 = \mathcal{A}\mathbf{\Pi}^0_1.$$

Proof. (1) $\Rightarrow$ (2): Take $f : \mathbb{N}^{\mathbb{N}} \to X$ continuous with $A = f(\mathbb{N}^{\mathbb{N}})$. Let $F_s := \overline{f([s])}$. Then Lemma 3.18 gives $A = \mathcal{A}(F_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$.

(2) $\Rightarrow$ (3): Trivial since closed sets are analytic.

(3) $\Rightarrow$ (1): By the observation above, $\mathcal{A}\mathbf{\Sigma}^1_1 \subseteq \mathbf{\Sigma}^1_1$.

For the last claim, note that $\mathcal{A}\mathbf{\Pi}^0_1 \subseteq \mathcal{A}\mathbf{\Sigma}^1_1$ since closed sets are analytic, while (1) $\Rightarrow$ (2) gives $\mathbf{\Sigma}^1_1 \subseteq \mathcal{A}\mathbf{\Pi}^0_1$ (with $\varnothing = \mathcal{A}(\varnothing)_{s\in\mathbb{N}^{<\mathbb{N}}}$) and the observation gives $\mathcal{A}\mathbf{\Sigma}^1_1 \subseteq \mathbf{\Sigma}^1_1$. Hence

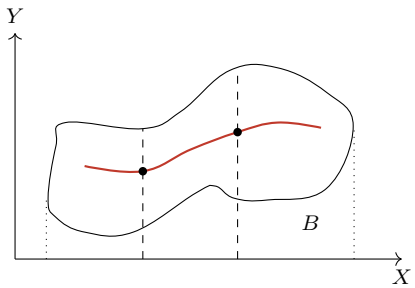
$$\mathbf{\Sigma}^1_1 \subseteq \mathcal{A}\mathbf{\Pi}^0_1 \subseteq \mathcal{A}\mathbf{\Sigma}^1_1 \subseteq \mathbf{\Sigma}^1_1,$$

so all three classes coincide; in particular $\mathcal{A}\mathbf{\Sigma}^1_1 = \mathcal{A}\mathbf{\Pi}^0_1$. □

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4 Uniformization Theorems

Uniformization theorems are about building nice right inverses to projections, i.e. given $B \subseteq X \times Y$, we'd like a way to uniformly pick a witness $y_x \in Y$ for each $x \in \mathrm{proj}_X(B)$ such that $(x, y_x) \in B$. In particular, given a Borel set $B \subseteq X \times Y$ in a product of Polish spaces, we ask when proj_X admits a Borel right inverse $f : \mathrm{proj}_X(B) \to B$. We will see that $\mathrm{proj}_X(B)$ is Borel in this case as a consequence.



It turns out that this is possible when the projection is "small"-to-one, e.g. one-to-one, countable-to-one, compact-to-one or K_σ (i.e. σ-compact)-to-one.

4.1 Projections and Compactness

Proposition 4.1

(a) Continuous functions map compact sets to compact sets.
(b) Continuous functions map K_σ sets to K_σ sets.
(c) (Tube lemma) For topological spaces X, Y, where Y is compact, $\mathrm{proj}_X : X \times Y \to X$ maps closed sets to closed sets.

Proof. (a), (b) are exercises ((a) is also Part 1, Theorem 3.7(d)).

(c) (HW 3) Let $F \subseteq X \times Y$ be closed. Then for $x \in X$, $x \notin \mathrm{proj}_X(F) \iff \forall y \in Y$, $(x, y) \in F^c$. Since F^c is open, for each $y \in Y$ there are open $U_y \ni x$, $V_y \ni y$ such that $(x, y) \in U_y \times V_y \subseteq F^c$. $\{V_y\}_{y \in Y}$ is an open cover of Y, hence there is a finite subcover $\{V_{y_1}, \ldots, V_{y_n}\}$ of Y. It is left as an exercise to check that $x \in \bigcap_{i=1}^n U_{y_i} \subseteq \mathrm{proj}_X(F)^c$. Hence $\mathrm{proj}_X(F)^c$ is open, thus

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$\mathrm{proj}_X(F)$ is closed. □

Example 4.2

(a) $\mathrm{proj}_0 : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ does not map closed sets to closed sets. E.g. $B := \mathrm{graph}\,(x \mapsto \frac{1}{x},\ x \neq 0)$ is closed in $\mathbb{R} \times \mathbb{R}$, but $\mathrm{proj}_0(B) = \mathbb{R} \setminus \{0\}$ is not closed.
(b) Because $\mathbb{R}$ is σ-compact, so is $\mathbb{R} \times \mathbb{R}$, hence every closed subset of $\mathbb{R} \times \mathbb{R}$ is σ-compact; and proj_0 maps σ-compact sets to σ-compact sets. So proj_0 maps closed sets to σ-compact (in particular F_σ) sets, e.g. $\mathrm{proj}_0(B) = \mathbb{R} \setminus \{0\} = \bigcup_{n \geq 1} \left([-n, -\frac{1}{n}] \cup [\frac{1}{n}, n]\right)$ in (a). □

4.2 The Luzin–Souslin Theorem

Theorem 4.3 (Luzin–Souslin; one-to-one)

Let X, Y be Polish, and $f : X \to Y$ be Borel with $f|_A$ injective for a Borel $A \subseteq X$. Then $f(A)$ is still Borel.

Proof (Ruiyuan Chen). Suppose first $X = A$ and f is continuous, so we need to show that $f(X) \subseteq Y$ is Borel. Let $\mathcal{T}_X, \mathcal{T}_Y$ denote the Polish topologies on X, Y, respectively. Fix a countable basis $\mathcal{U} \subseteq \mathcal{T}_X$ for $\mathcal{T}_X$. For $U \in \mathcal{U}$, by injectivity, $f(U)$ and $f(U^c)$ are disjoint (and analytic), thus by Luzin's analytic separation, there is a Borel set $B_U \subseteq Y$ such that $B_U \supseteq f(U)$ and $B_U^c \supseteq f(U^c)$. Let $\mathcal{T}'_Y \supseteq \mathcal{T}_Y$ be a Polish topology in which all the B_U are clopen, with $\mathcal{B}(\mathcal{T}'_Y) = \mathcal{B}(\mathcal{T}_Y)$. We now show that $f(X)$ is G_δ in $\mathcal{T}'_Y$; then $f(X) \in \mathcal{B}(\mathcal{T}'_Y) = \mathcal{B}(\mathcal{T}_Y)$.

Notice $\mathrm{graph}(f) \subseteq X \times Y$ is closed in $\mathcal{T}_X \times \mathcal{T}_Y$, hence closed in $\mathcal{T}_X \times \mathcal{T}'_Y$ by refinement. So by making closed-graph functions continuous, the topology $\mathcal{T}'_X := \mathcal{T}_{X,f}$ generated by $f : X \to (Y, \mathcal{T}'_Y)$ over $\mathcal{T}_X$ is still Polish, and $f : (X, \mathcal{T}'_X) \to (Y, \mathcal{T}'_Y)$ is continuous. In fact, f is an open map onto $f(X)$: $\mathcal{T}'_X$ is

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generated by $\mathcal{U} \cup \{f^{-1}(V) : V \in \mathcal{T}'_Y\}$, so every $\mathcal{T}'_X$-open set is a union of sets $U \cap f^{-1}(V)$ with $U \in \mathcal{U}$, $V \in \mathcal{T}'_Y$ (the observation after Definition 2.12). Moreover $f(f^{-1}(V)) = V \cap f(X)$, and $f(U) = B_U \cap f(X)$ by the choice of B_U (and injectivity), thus

$$f(U \cap f^{-1}(V)) = B_U \cap f(X) \cap V$$

is open in $f(X)$. Hence f is an embedding, so $f(X) \cong (X, \mathcal{T}'_X)$ is Polish, therefore G_δ in $(Y, \mathcal{T}'_Y)$. Thus $f(X) \in \mathcal{B}(\mathcal{T}_Y)$.

For general $A \subseteq X$ and Borel f: since the assertion is about $f(A)$ in $\mathcal{T}_Y$, we can safely refine $\mathcal{T}_X$ so that f is continuous and A is clopen (clopenify A and the sets $f^{-1}(V)$ for V in a countable basis of Y). Then applying what we just proved to the Polish space A and the continuous injection $f|_A$ gives the desired result. □

Corollary 4.4

(a) If X is a set with Polish topologies $\mathcal{T}_0, \mathcal{T}_1$ on it and $\mathcal{T}_0 \subseteq \mathcal{B}(\mathcal{T}_1)$, then $\mathcal{B}(\mathcal{T}_0) = \mathcal{B}(\mathcal{T}_1)$.

(b) A subset B of a Polish space X is Borel $\iff$ B is a continuous injective image of a closed subset of $\mathbb{N}^{\mathbb{N}}$.

(c) Let X, Y be Polish and $G \subseteq X \times Y$ be a Borel function graph, i.e. $G \in \mathcal{B}(X \times Y)$ such that $\forall x \in X$, $\exists^{\leq 1}$ (there exists at most one) $y \in Y$ with $(x, y) \in G$. In other words, G is the graph of a partial function $f : X \rightharpoonup Y$. Then $\mathrm{dom}(f) \subseteq X$ is Borel.

Proof. (a) Since $\mathcal{T}_0 \subseteq \mathcal{B}(\mathcal{T}_1)$, $\langle \mathcal{T}_0 \rangle_\sigma = \mathcal{B}(\mathcal{T}_0) \subseteq \mathcal{B}(\mathcal{T}_1)$. Consider the injective Borel map $\mathrm{id} : (X, \mathcal{T}_1) \to (X, \mathcal{T}_0)$. Then by Luzin–Souslin, $\forall B \in \mathcal{B}(\mathcal{T}_1)$, $\mathrm{id}(B) = B$ is $\mathcal{T}_0$-Borel. Hence

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$\mathcal{B}(\mathcal{T}_1) = \mathcal{B}(\mathcal{T}_0)$.

(b) By clopenification, ($\Rightarrow$) is known. ($\Leftarrow$) is immediate from the Luzin–Souslin theorem.

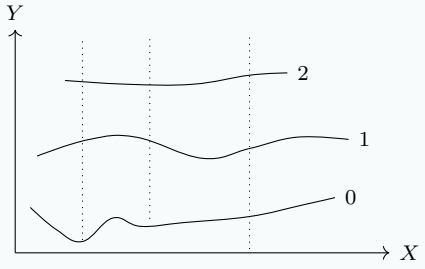
(c) $\mathrm{dom}(f) = \mathrm{proj}_X(G)$, which is Borel by Luzin–Souslin since $\mathrm{proj}_X|_G$ is injective. □

4.3 The Luzin–Novikov Theorem and Countable Uniformization

One of the most important theorems in DST is the following.

Theorem 4.5 (Luzin–Novikov)

Let X, Y be Polish spaces and $R \subseteq X \times Y$ be Borel such that every X-fiber of R is countable, i.e. R_x is countable $\forall x \in X$. Then $R = \bigsqcup_{n \in \mathbb{N}} G_n$ for some Borel function graphs G_n; equivalently, $G_n = \mathrm{graph}(\gamma_n)$ for Borel $\gamma_n : B_n \to Y$, with B_n Borel in X.



Proof (Forte Shinko). Deferred to one of the final presentations. □

Corollary 4.6

Countable-to-one Borel images of Borel sets are Borel, i.e. if X, Y are Polish, $B \subseteq X$ is Borel and $f : X \to Y$ is Borel such that $f|_B$ is countable-to-one, then $f(B)$ is Borel.

Proof. Recall that for a map f between Polish spaces, f is Borel $\iff$ graph(f) is a Borel subset of $X \times Y$. Hence $C := \{(f(x), x) : x \in B\} \subseteq Y \times X$, the image of graph$(f) \cap (B \times Y)$ under the swap homeomorphism, is Borel, and its Y-fibers $C_y = \{x \in B : f(x) = y\}$ are countable. So it suffices to show that if $C \subseteq Y \times X$ is Borel with countable

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Y-fibers, then $\mathrm{proj}_Y(C)$ $(= f(B))$ is Borel. Yet by the Luzin–Novikov theorem, $C = \bigsqcup_{n\in\mathbb{N}} C_n$, with each C_n a Borel function graph. But then $\mathrm{proj}_Y(C) = \bigcup_{n\in\mathbb{N}} \mathrm{proj}_Y(C_n)$, with each $\mathrm{proj}_Y(C_n)$ Borel by Luzin–Souslin. □

Corollary 4.7 (Countable uniformization; an equivalent form of Luzin–Novikov)

For Polish spaces X, Y, every countable-to-one Borel function $f : X \to Y$ admits a Borel right inverse $g : f(X) \to X$, with $f(X)$ Borel.

Proof. Let $G := \mathrm{graph}(f)^{-1} := \{(y, x) \in Y \times X : (x, y) \in \mathrm{graph}(f)\}$, a Borel set; thus G has countable Y-fibers since f is countable-to-one. By Luzin–Novikov, $G = \bigsqcup_{n\in\mathbb{N}} G_n$, each G_n a Borel function graph. Define

$$\begin{aligned} k : f(X) &\to \mathbb{N}, & y &\mapsto \min\{n \in \mathbb{N} \mid y \in \mathrm{proj}_Y G_n\}, \\ g : f(X) &\to X, & y &\mapsto \text{the unique } x \in X \text{ such that } (y, x) \in G_{k(y)}. \end{aligned}$$

Then k is Borel: it suffices to check this with the singletons. For every $n \in \mathbb{N}$, $k^{-1}(\{n\}) = \mathrm{proj}_Y G_n \setminus \bigcup_{i<n} \mathrm{proj}_Y(G_i)$, with each $\mathrm{proj}_Y G_i$ Borel by Luzin–Souslin. Moreover, $f(X) = \bigcup_n \mathrm{proj}_Y G_n$ is Borel.

Lastly, we show g is Borel. Pick $U \subseteq X$ open, then

$$\begin{aligned} y \in g^{-1}(U) &\iff g(y) \in U \\ &\iff \exists\, x \in U \text{ such that } (y, x) \in G_{k(y)} \\ &\iff \exists\, n \in \mathbb{N} \text{ such that } y \in k^{-1}(\{n\}) \ \wedge\ \exists\, x \in U \text{ such that } (y, x) \in G_n \\ &\iff \underbrace{\exists\, n \in \mathbb{N}}_{\text{countable } \cup} \text{ such that } \underbrace{y \in k^{-1}(\{n\})}_{\text{Borel}} \wedge \underbrace{y \in \mathrm{proj}_Y\left(G_n \cap (Y \times U)\right)}_{\text{Borel}}, \end{aligned}$$

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where $\mathrm{proj}_Y\left(G_n \cap (Y \times U)\right)$ is Borel by Corollary 4.4(c), as $G_n \cap (Y \times U)$ is still a Borel function graph. Thus $f \circ g = \mathrm{id}_{f(X)}$ with g Borel. □

4.4 Countable Borel Equivalence Relations

The following notion is why DST is applicable in dynamics and measured group theory.

Definition 4.8

A countable Borel equivalence relation (CBER) is an equivalence relation E on a Polish space X such that E is Borel as a subset of $X \times X$ and each E-class is countable. □

Example 4.9

If Γ is a countable group and $\Gamma \curvearrowright X$ a Borel action on a Polish space X, i.e. each $\gamma \in \Gamma$ acts as a Borel automorphism of X, then the orbit equivalence relation E_Γ of this action is a CBER, since Γ is countable and by definition of a group action, for $x, y \in X$,

$$x \, E_\Gamma \, y \iff \underbrace{\exists \gamma \in \Gamma}_{\text{countable } \cup} \; \underbrace{(\gamma x = y)}_{\text{Borel}},$$

where $\{(x,y) : \gamma x = y\} = \operatorname{graph}(\gamma)$ is Borel, being the graph of the Borel function $\gamma : X \to X$. For example, take the shift action $\Gamma \curvearrowright 2^\Gamma$, where

$$(\gamma \cdot x)(\delta) := x(\delta\gamma), \qquad \delta, \gamma \in \Gamma, \; x \in 2^\Gamma. \quad \square$$

The following theorem says these are the only examples of CBERs.

Corollary 4.10 (Feldman–Moore theorem)

Every CBER $E \subseteq X^2$ on a Polish space X is a countable union of graphs of Borel involutions on X, i.e. $E = \bigcup_{n \in \mathbb{N}} \operatorname{graph}(\gamma_n)$, where each $\gamma_n : X \to X$ is

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a Borel involution, i.e. $\gamma_n^{-1} = \gamma_n$. In particular, E is the orbit equivalence relation of a Borel action of a countable group Γ generated by involutions. In fact,

$$E \setminus \Delta(X) = \bigsqcup_{n \in \mathbb{N}} \left(\operatorname{graph}(\gamma_n) \setminus \Delta(X) \right), \qquad \text{where } \Delta(X) := \{(x,x) \mid x \in X\}. \tag{$\dagger$}$$

Proof. We do the classical proof; a proof with Borel edge-colouring is carried out in HW 4.

Convention. For $R \subseteq X^2$, we denote $R^{-1} := \{(y,x) : (x,y) \in R\}$. We view a Borel (partial) function as both a function and its graph, interchangeably: a Borel function graph $f \subseteq X^2$ is at the same time the partial function $f : X \rightharpoonup X$ with domain $\operatorname{dom}(f) := \operatorname{proj}_0(f)$ and range $\operatorname{ran}(f) := \operatorname{proj}_1(f)$, where $\operatorname{proj}_0, \operatorname{proj}_1 : X^2 \to X$ are the projections onto the first and the second coordinate; so $(x,y) \in f$ means $f(x) = y$. When the f_n are written in a union (or an intersection, or a difference), that means taking the union of their graphs in X^2. Sometimes, when we need to emphasize that we are thinking of the graph, we write $\operatorname{graph}(f_n)$. In particular, f^{-1} always denotes the inverse relation; when f is injective, it is (the graph of) the inverse function.

By Corollary 4.4(c), $\operatorname{dom}(f)$ is Borel and f is a Borel function on it. Its range $\operatorname{ran}(f)$ is analytic in general, but it is Borel if f is injective, by Luzin–Souslin again.

Step 1 (injective pieces). By Luzin–Novikov (each fiber E_x is an E-class, hence countable), we can write E as a disjoint union of Borel partial functions, $E = \bigsqcup_{n \in \mathbb{N}} f_n$. We will use these partial functions

to build a set of Borel involutions of X. Notice

$$E = E^{-1} = \bigsqcup_{m\in\mathbb{N}} f_m^{-1}$$

since E is symmetric, where f_m^{-1} is the inverse relation of f_m (the inverse function need not exist yet). One can verify that the partial function $f_n \cap f_m^{-1}$ is injective and Borel: if $(x,y),(x',y) \in f_n \cap f_m^{-1}$, then $(y,x),(y,x') \in f_m$, so $x = f_m(y) = x'$. Since $E = E \cap E^{-1} = \bigsqcup_{n,m\in\mathbb{N}}(f_n \cap f_m^{-1})$, WLOG we

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replace $\{f_n\}_n$ by $\{f_n \cap f_m^{-1}\}_{n,m}$ (re-indexed by $\mathbb{N}$) and assume the f_n's are injective with $E = \bigsqcup_{n\in\mathbb{N}} f_n$.

Step 2 (disjoint domains and ranges). Next, we want to extend each f_n to a Borel involution of X. If $\operatorname{dom}(f_n) \cap \operatorname{ran}(f_n) = \varnothing$, then this can be easily done:

$$\tilde{f}_n(x) := \begin{cases} f_n(x) & \text{if } x \in \operatorname{dom}(f_n), \\ f_n^{-1}(x) & \text{if } x \in \operatorname{ran}(f_n), \\ x & \text{otherwise,} \end{cases} \qquad \text{for } n \in \mathbb{N}. \tag{$*$}$$

Then this is clearly an involution, and it is Borel since f_n, f_n^{-1} and $x \mapsto x$ are Borel with Borel domains.

We now make $\operatorname{dom}(f_n)$ and $\operatorname{ran}(f_n)$ disjoint as follows. Pick a countable basis $\mathcal{W} = (W_k)_{k\in\mathbb{N}}$ of X; thus $\mathcal{W}^2 := (W_k \times W_l)_{k,l\in\mathbb{N}}$ is a countable basis of X^2. Let $(x,y) \in X^2$ with $x \neq y$. Then there are disjoint basic open sets $U_{(x,y)} \ni x$, $V_{(x,y)} \ni y$ in $\mathcal{W}$, hence $U_{(x,y)} \times V_{(x,y)} \in \mathcal{W}^2$ and $(x,y) \in U_{(x,y)} \times V_{(x,y)} \subseteq X^2 \setminus \Delta(X)$ by disjointness. Hence

$$X^2 \setminus \Delta(X) = \bigcup_{(x,y)\notin\Delta(X)} \underbrace{U_{(x,y)} \times V_{(x,y)}}_{\in \mathcal{W}^2}.$$

But $\mathcal{W}^2$ is countable, thus the union is countable: $X^2 \setminus \Delta(X) = \bigcup_{m\in\mathbb{N}} U_m \times V_m$ with $U_m \cap V_m = \varnothing$. Hence

$$E = \Delta(X) \cup \Big(E \cap \bigcup_m (U_m \times V_m)\Big) = \Delta(X) \cup \bigcup_{n,m\in\mathbb{N}} (f_n \cap (U_m \times V_m)).$$

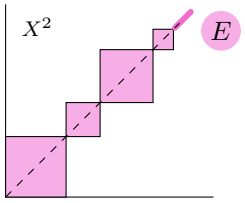
Every graph in the latter union has its domain ($\subseteq U_m$) and range ($\subseteq V_m$) disjoint. Again, replacing $\{f_n\}_{n\in\mathbb{N}}$ with $\{f_n \cap (U_m \times V_m)\}_{n,m\in\mathbb{N}}$, we assume each Borel injective f_n has disjoint domain and range, and now

$$E = \Delta(X) \sqcup \bigcup_{n\in\mathbb{N}} f_n.$$

(The union of the f_n's need not be disjoint any more; the union with $\Delta(X)$ is, since every f_n misses the diagonal.)

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Step 3 (the involutions; $E = E_\Gamma$).



Extend each f_n into a Borel involution $\tilde{f}_n =: \gamma_n$ as in $(*)$. Let Γ be the group generated by $(\gamma_n)_n$ under composition. Then the natural action $\Gamma \curvearrowright X$ is Borel. Furthermore, since E is reflexive and symmetric, $\operatorname{graph}(\gamma_n) = f_n \cup f_n^{-1} \cup \{(x,x) : x \notin \operatorname{dom}(f_n) \cup \operatorname{ran}(f_n)\} \subseteq E$, so by our construction

$$E = \Delta(X) \cup \bigcup_{n\in\mathbb{N}} \operatorname{graph}(f_n) = \Delta(X) \cup \bigcup_{n\in\mathbb{N}} \operatorname{graph}(\gamma_n),$$
$$\text{and} \quad E_\Gamma = \bigcup_{\gamma\in\Gamma} \{(x,\gamma(x)) : x \in X\} = \bigcup_{\gamma\in\Gamma} \operatorname{graph}(\gamma).$$

Lastly, we show $E = E_\Gamma$. Clearly $E \subseteq E_\Gamma$ since $(\gamma_n)_n$ generates Γ and $\Delta(X) = \operatorname{graph}(\mathrm{id}_X)$ for $\mathrm{id}_X \in \Gamma$. Conversely, take an element $\gamma = \gamma_{n_1} \circ \gamma_{n_2} \circ \cdots \circ \gamma_{n_k} \in \Gamma$ (every element has this form, as each generator is its own inverse). Then for any $x \in X$, γ acts on x by

$$x \overset{\gamma_{n_k}}{\longmapsto} \gamma_{n_k}(x) \longmapsto \cdots \overset{\gamma_{n_1}}{\longmapsto} \gamma(x).$$

But each step is an equivalence in E since $\operatorname{graph}(\gamma_n) \subseteq E$ for the generators, thus by transitivity of E, $x \mathrel{E} \gamma(x)$. Thus every pair $(x,\gamma(x)) \in E_\Gamma$ is in E, hence $E_\Gamma = E$.

Step 4 (the disjoint version (†)). It remains to show (†). However, with our construction of the γ_n and the symmetry of E together, we can easily see that $\bigcup_n (\operatorname{graph}(\gamma_n) \setminus \Delta(X))$ can never be a disjoint union (unless $E = \Delta(X)$): if $(x,y) \in f_n$, then $(y,x) \in E \setminus \Delta(X)$ lies in some f_m, with $m \neq n$ as $y \notin \operatorname{dom}(f_n)$; so $(y,x) \in \operatorname{graph}(\gamma_n) \cap \operatorname{graph}(\gamma_m)$. We fix it as follows. Before turning f_n into γ_n, we have $E = \Delta(X) \sqcup \bigcup_n f_n$ (Step 2) with each f_n a Borel partial function that is injective and $\operatorname{dom}(f_n) \cap \operatorname{ran}(f_n) = \varnothing$. Thus $E \setminus \Delta(X) = \bigcup_{n\in\mathbb{N}} f_n$. We now do the same trick as in countable uniformization: set

$$c : E \setminus \Delta(X) \to \mathbb{N}, \qquad (x,y) \mapsto \text{the least } n \in \mathbb{N} \text{ such that } (x,y) \in f_n.$$

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c is Borel since $c^{-1}(\{n\}) = f_n \setminus \bigcup_{i<n} f_i$. Symmetrize c by $k(x,y) := \min\{c(x,y), c(y,x)\}$. The map $(x,y) \mapsto c(y,x)$ is c composed with the swap homeomorphism, hence Borel; thus k is Borel. Hence each class $M_n := k^{-1}(\{n\})$ is a symmetric Borel subset of $E \setminus \Delta(X)$. Moreover, for all $x \in X$, there are no $y \neq z$ with $(x,y),(x,z) \in M_n$: if not, then by definition of k and M_n,

$$n = \begin{Bmatrix} c(x,y) \\ \text{or} \\ c(y,x) \end{Bmatrix} = \begin{Bmatrix} c(x,z) \\ \text{or} \\ c(z,x) \end{Bmatrix}.$$

All 4 cases are impossible:

- $c(x,y) = c(x,z) = n$ means $(x,y),(x,z) \in \operatorname{graph}(f_n)$, i.e. $f_n(x) = y$ and $f_n(x) = z$ with $y \neq z$, impossible;
- $(y,x),(z,x) \in \operatorname{graph}(f_n)$ is impossible since f_n is injective;
- $(y,x),(x,z) \in \operatorname{graph}(f_n)$ or $(x,y),(z,x) \in \operatorname{graph}(f_n)$ are impossible since $\operatorname{dom}(f_n) \cap \operatorname{ran}(f_n) = \varnothing$.

Lastly, we redefine the involutions: let $\gamma_n(x) :=$ the unique y with $(x,y) \in M_n$ if it exists, else x. It is an involution by symmetry of M_n, and Borel since

$$\operatorname{graph}(\gamma_n) = M_n \cup \{(x,x) : x \notin \operatorname{proj}_0(M_n)\}$$

is Borel in X^2 ($\mathrm{proj}_0(M_n)$ is Borel by Corollary 4.4(c)). Now $\mathrm{graph}(\gamma_n) \setminus \Delta(X) = M_n$; $E_\Gamma = E$ for the group Γ generated by these γ_n, in the same way as before (Step 3: $\mathrm{graph}(\gamma_n) \subseteq E$ and $E = \Delta(X) \cup \bigsqcup_n M_n$); and

$$E \setminus \Delta(X) = \bigcup_{n\in\mathbb{N}} (\mathrm{graph}(f_n) \setminus \Delta(X)) = \bigsqcup_{n\in\mathbb{N}} M_n = \bigsqcup_{n\in\mathbb{N}} (\mathrm{graph}(\gamma_n) \setminus \Delta(X)).$$ □

4.5 Further Uniformization Theorems

Lastly, we have the most powerful uniformization theorems available, which we won't prove.

Theorem 4.11 (Arsenin–Kunugui)

Let X, Y be Polish and $B \subseteq X \times Y$ be

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Borel with every X-fiber B_x being K_σ. Then

(1) $\mathrm{proj}_X(B)$ is Borel (analog of Luzin–Souslin);
(2) $\mathrm{proj}_X|_B$ has a Borel right inverse $\mathrm{proj}_X(B) \to B$ (analog of countable uniformization).

Theorem 4.12 (Saint Raymond)

Let X, Y be Polish and $B \subseteq X \times Y$ be Borel with every X-fiber B_x being K_σ. Then $B = \bigcup_{n\in\mathbb{N}} B_n$, where each B_n is Borel and every X-fiber $(B_n)_x$ is compact (analog of the Luzin–Novikov theorem).

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5 Standard Borel Spaces

5.1 The Borel Isomorphism Theorem

We will prove that any two uncountable Polish spaces are Borel isomorphic.

Corollary 5.1 (Borel Cantor–Schröder–Bernstein)

Let X, Y be Polish and $f : X \hookrightarrow Y$, $g : Y \hookrightarrow X$ be Borel injections. Then X, Y are Borel isomorphic.

Proof. The usual proof for CSB (monotone map → fixed set → bijection) fails to show Borel-ness. We implement a fix to it instead of doing the usual Knaster–Tarski construction.

For $A \subseteq X$, put $\varphi(A) := X \setminus g(Y \setminus f(A))$. Then $\varphi : \mathcal{P}(X) \to \mathcal{P}(X)$ is monotone, i.e. for $A \subseteq B \subseteq X$, $\varphi(A) \subseteq \varphi(B)$.

Furthermore, φ preserves Borel sets: let A be Borel. Then $f(A)$ is Borel by Luzin–Souslin, and so is $Y \setminus f(A)$. Thus $g(Y \setminus f(A))$ is Borel by Luzin–Souslin again, and so is its complement $\varphi(A)$.

Lastly, φ commutes with countable intersections: first notice that $f(\bigcap_n A_n) = \bigcap_n f(A_n)$ (exercise: use injectivity of f). Then, since images commute with unions,

$$\begin{aligned}\varphi\Big(\bigcap_n A_n\Big) &= X \setminus g\Big(Y \setminus f\Big(\bigcap_n A_n\Big)\Big) = X \setminus g\Big(Y \setminus \bigcap_n f(A_n)\Big)\\ &= X \setminus g\Big(\bigcup_n (Y \setminus f(A_n))\Big) = X \setminus \bigcup_n g(Y \setminus f(A_n))\\ &= \bigcap_n \Big(X \setminus g(Y \setminus f(A_n))\Big) = \bigcap_n \varphi(A_n).\end{aligned}$$

Let $X_n := \varphi^n(X)$. Then $(X_n)_n$ is decreasing: $X_1 \subseteq X = X_0$, and then by monotonicity, $X_2 = \varphi(X_1) \subseteq \varphi(X_0) = X_1$, and the

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rest follows by induction. Since φ preserves Borel sets, all the X_n are Borel by induction. Put $A_0 := \bigcap_n X_n$, which is Borel as a countable intersection of Borel sets.

Now observe that A_0 is a fixed point: since φ commutes with countable intersections,

$$\varphi(A_0) = \bigcap_n \varphi(X_n) = \bigcap_n X_{n+1} = A_0$$

($\bigcap_n X_n = \bigcap_n X_{n+1}$ since the X_n decrease). What remains is the same construction as in the usual CSB: define $h : X \to Y$ by

$$h(x) := \begin{cases} f(x), & x \in A_0,\\ g^{-1}(x), & x \in X \setminus A_0.\end{cases}$$

This is clearly well-defined and bijective.

Let $C \subseteq Y$ be Borel. Then notice that

$$h^{-1}(C) = (A_0 \cap f^{-1}(C)) \cup g(C \setminus f(A_0)),$$

where A_0 is Borel as discussed, $f^{-1}(C)$ is Borel by assumption, and $f(A_0)$ is Borel by Luzin–Souslin, hence $g(C \setminus f(A_0))$ is Borel by Luzin–Souslin again. Thus $h^{-1}(C)$ is Borel, hence h is Borel. Given a Borel bijection, its inverse is immediately Borel by Luzin–Souslin again. Hence X is Borel isomorphic to Y by h. □

Theorem 5.2 (Borel Isomorphism Theorem)

Any two uncountable Polish spaces are Borel isomorphic.

Proof. Let X be an uncountable Polish space. By Cantor–Bendixson (together with Part 1, Theorem 4.5), there is a topological embedding $f : 2^{\mathbb{N}} \hookrightarrow X$; and by binary representation, there is an injection $b : X \hookrightarrow 2^{\mathbb{N}}$ such that for the

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clopen sets $[n \mapsto 1] := \{x \in 2^{\mathbb{N}} : x(n) = 1\}$, $b^{-1}([n \mapsto 1])$ is open. It is obvious that for the basic open cylinder sets U in $2^{\mathbb{N}}$, $b^{-1}(U)$ is Borel, hence b is Borel. Thus by Borel CSB, X is Borel isomorphic to $2^{\mathbb{N}}$. □

Definition 5.3

A measurable space $(X, \mathcal{S})$ is called a standard Borel space if $\mathcal{S} = \mathcal{B}(\mathcal{T})$ for some Polish topology $\mathcal{T}$ on X. □

Corollary 5.4

There is a unique uncountable standard Borel space up to Borel isomorphism, namely $2^{\mathbb{N}}$.

Proof. Apply the Borel Isomorphism Theorem. □

Example 5.5

(a) For a Polish space X, $(X, \mathcal{B}(X))$ is standard Borel.

(b) Let X be Polish and $B \subseteq X$ be Borel. Then $(B, \mathcal{B}(X)|_B)$ is standard Borel, where $\Gamma(X)|_B := \{C \cap B : C \in \Gamma(X)\}$. In particular, since B is Borel,

$$\mathcal{B}(X)|_B = \{C \cap B : C \subseteq X \text{ Borel}\} = \{C \in \mathcal{B}(X) : C \subseteq B\}. \quad \square$$

Proof of (b). Let $\mathcal{T}$ be the Polish topology on X and $\mathcal{T}' \supseteq \mathcal{T}$ be a Polish topology making B clopen with $\mathcal{B}(\mathcal{T}') = \mathcal{B}(\mathcal{T})$. Then $(B, \mathcal{T}'|_B)$ is Polish, so $(B, \mathcal{B}(\mathcal{T}'|_B))$ is standard Borel. But $\mathcal{B}(\mathcal{T}'|_B) = \mathcal{B}(\mathcal{T}')|_B = \mathcal{B}(\mathcal{T})|_B$. □

Proposition 5.6

Example 5.5(b) characterizes Borel sets, i.e. if X is Polish, $A \subseteq X$ and $(A, \mathcal{B}(X)|_A)$ is standard Borel, then $A \subseteq X$ is Borel.

Proof. Note that this together with Example 5.5(b) gives HW 4, Q6.

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By standard-Borel-ness, there is a Polish topology $\mathcal{T}$ on A such that $\mathcal{B}(\mathcal{T}) = \mathcal{B}(X)|_A$. Define $\iota : (A, \mathcal{T}) \hookrightarrow X$ by the identity map. For Borel $C \subseteq X$,

$$\iota^{-1}(C) = C \cap A \in \mathcal{B}(X)|_A = \mathcal{B}(\mathcal{T}).$$

Thus ι is a Borel injection, and A is Borel in $\mathcal{B}(X)|_A = \mathcal{B}(\mathcal{T})$. Hence $\iota(A) = A$ is Borel in X by the Luzin–Souslin theorem. □

5.2 The Effros Borel Space

Definition 5.7

For a Polish space X, let $\mathcal{F}(X)$ denote the set of all closed subsets of X. The Effros Borel space of X is the measurable space $(\mathcal{F}(X), \mathcal{E})$, where $\mathcal{E}$ is the σ-algebra generated by the sets of the form

$$[U] := \{F \in \mathcal{F}(X) : F \cap U \neq \varnothing\}, \quad \text{for all open } U \subseteq X. \quad \square$$

Theorem 5.8

For every Polish space X, its Effros Borel space is standard Borel.

Proof. We use binary representation again. Let $(U_n)_{n\in\mathbb{N}}$ be a basis of nonempty open sets for the Polish topology on X, and define $c : \mathcal{F}(X) \to 2^{\mathbb{N}}$ by $F \mapsto (F \cap U_n \neq \varnothing)_{n\in\mathbb{N}}$, i.e. the n-th bit is 1 if $F \cap U_n \neq \varnothing$, else 0.

This map is injective: if $F_0 \neq F_1$ are closed sets in X, then at least one of $F_0 \setminus F_1$ and $F_1 \setminus F_0$ is nonempty, say $F_0 \setminus F_1 \neq \varnothing$. Then for all $x \in F_0 \setminus F_1$, there is a U_n such that $x \in U_n \subseteq F_1^c$, i.e. $U_n \cap F_1 = \varnothing$. Hence $c(F_0)(n) = 1$ and $c(F_1)(n) = 0$, i.e. $c(F_0) \neq$

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$c(F_1)$.

We verify that c is Borel. Notice that the cylinders $([n \mapsto 1])_{n\in\mathbb{N}}$ generate $\mathcal{B}(2^{\mathbb{N}})$. Hence it suffices to check that $c^{-1}([n \mapsto 1])$ is Borel for every $n \in \mathbb{N}$. But

$$c^{-1}([n \mapsto 1]) = \{F \subseteq X : F \text{ closed}, \ F \cap U_n \neq \varnothing\} = [U_n],$$

which is Effros Borel by definition.

Furthermore, c maps Effros Borel sets to Borel sets relative to the image $c(\mathcal{F}(X))$: the $[U_n]$'s generate the Effros Borel σ-algebra since for every open $U \subseteq X$, $[U] = \bigcup_{U_n \subseteq U} [U_n]$. The map c commutes with arbitrary unions, and it commutes with taking complements relative to $c(\mathcal{F}(X))$. Lastly, notice that $c([U_n]) = [n \mapsto 1] \cap c(\mathcal{F}(X))$. Hence c maps the generating sets into the Borel σ-algebra $\mathcal{B}(2^{\mathbb{N}})|_{c(\mathcal{F}(X))}$, thus c maps all the Effros Borel sets to Borel sets in $\mathcal{B}(2^{\mathbb{N}})|_{c(\mathcal{F}(X))}$, as c commutes with complements and intersections.

Denote the image $c(\mathcal{F}(X)) =: \mathcal{C}$. We have shown that $(\mathcal{F}(X), \mathcal{E})$ is Borel isomorphic to $(\mathcal{C}, \mathcal{B}(2^{\mathbb{N}})|_{\mathcal{C}})$. If $\mathcal{C}$ is a Borel subset of $2^{\mathbb{N}}$, then by Example 5.5(b), $(\mathcal{C}, \mathcal{B}(2^{\mathbb{N}})|_{\mathcal{C}})$ is standard Borel, say $\mathcal{B}(2^{\mathbb{N}})|_{\mathcal{C}} = \mathcal{B}(\mathcal{T}_{\mathcal{C}})$ for a Polish topology $\mathcal{T}_{\mathcal{C}}$ on $\mathcal{C}$; and with $\mathcal{T} := \{c^{-1}(V) : V \in \mathcal{T}_{\mathcal{C}}\}$, $(\mathcal{F}(X), \mathcal{T})$ is homeomorphic to $(\mathcal{C}, \mathcal{T}_{\mathcal{C}})$, hence Polish, and $\mathcal{B}(\mathcal{T}) = c^{-1}(\mathcal{B}(2^{\mathbb{N}})|_{\mathcal{C}}) = \mathcal{E}$, since c is a

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Borel isomorphism.

Hence it remains to show $\mathcal{C} \in \mathcal{B}(2^{\mathbb{N}})$. To do so, it is HW 4, Q7 to check the following. Fix a complete

compatible metric d on X. Then for all $y \in 2^{\mathbb{N}}$,

$$\begin{aligned} y \in c(\mathcal{F}(X)) &\iff y = c(F) \text{ for some } F \in \mathcal{F}(X) \\ &\overset{\circledast}{\iff} (\forall n, m \in \mathbb{N} \text{ with } U_n \subseteq U_m,\ y(n) = 1 \Rightarrow y(m) = 1) \wedge \qquad (\dagger) \\ &\quad [\forall U_n\ \forall \varepsilon \in \mathbb{Q}^+,\ y(n) = 1 \Rightarrow (\exists \overline{U_m} \subseteq U_n \text{ with } \operatorname{diam}(U_m) < \varepsilon,\ y(m) = 1)] \qquad (*) \\ &\iff \underbrace{[\forall n, m \in \mathbb{N} \text{ such that } U_n \subseteq U_m,\ \underbrace{(y(n) = 0 \ \vee\ y(m) = 1)}_{\text{clopen}}]}_{\text{closed}} \wedge \\ &\quad \underbrace{[\forall U_n\ \forall \varepsilon \in \mathbb{Q}^+,\ \underbrace{y(n) = 0}_{\text{clopen}} \vee \underbrace{(\exists \overline{U_m} \subseteq U_n \text{ such that } \operatorname{diam}(U_m) < \varepsilon \wedge \underbrace{y(m) = 1}_{\text{clopen}})}_{\text{open}}]}_{G_\delta}. \end{aligned}$$

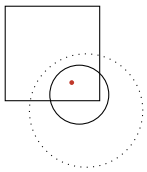
So $\mathcal{C}$ is the intersection of a closed set and a G_δ set, hence G_δ, in particular Borel. (So $\mathcal{C}$ is even Polish in the subspace topology by Part 1, Theorem 1.9, and we may take $\mathcal{T}_{\mathcal{C}}$ to be the subspace topology, i.e. $\mathcal{T} = \{c^{-1}(V \cap \mathcal{C}) : V \subseteq 2^{\mathbb{N}} \text{ open}\}$.)

We show the key equivalence $\circledast$.

($\Rightarrow$) If $y = c(F)$, then if $U_n \subseteq U_m$, $y(n) = 1 \Rightarrow y(m) = 1$ because $\varnothing \neq U_n \cap F \subseteq U_m \cap F$. For U_n and $\varepsilon \in \mathbb{Q}^+$, if $y(n) = 1$, then $U_n \cap F \neq \varnothing$. Let $x \in U_n \cap F$. Then there is $0 < \delta < \varepsilon/2$ with $B_\delta(x) \subseteq U_n$. Then, same as in the proof of the Baire category theorem, there is a U_m such that

$$x \in U_m \subseteq \overline{U_m} \subseteq B_{\delta/2}[x] \subseteq B_\delta(x) \subseteq U_n,$$

and $\operatorname{diam}(U_m) \leq \operatorname{diam}(B_\delta(x)) \leq 2\delta < \varepsilon$, with $y(m) = 1$.



($\Leftarrow$) We want to show that there is a closed F such that $F \cap U_n = \varnothing$ iff $y(n) = 0$. Set

$$F := X \setminus \bigcup_{\substack{n \in \mathbb{N} \\ y(n)=0}} U_n,$$

then F is closed, and $y(n) = 0 \Rightarrow F \cap U_n = \varnothing$. On the other hand, suppose $y(n) = 1$. By $(*)$, we

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construct a sequence

$$U_n \supseteq \overline{U_{m_0}} \supseteq U_{m_0} \supseteq \overline{U_{m_1}} \supseteq U_{m_1} \supseteq \overline{U_{m_2}} \supseteq \cdots,$$

with $y(m_j) = 1$ for all j and $\operatorname{diam}(U_{m_j}) = \operatorname{diam}(\overline{U_{m_j}}) < 1/j$ for $j \in \mathbb{N}^+$. By HW 1, Q2,

$$\bigcap_{j \in \mathbb{N}}^{\downarrow} U_{m_j} = \bigcap_{j \in \mathbb{N}}^{\downarrow} \overline{U_{m_j}} = \{x\}$$

for some $x \in X$. Now $x \in F$: if $x \in U_{\tilde{n}}$ with $y(\tilde{n}) = 0$, then $\bigcap^{\downarrow}_{j \in \mathbb{N}} U_{m_j} \subseteq U_{\tilde{n}}$ gives $j \in \mathbb{N}$ such that $U_{m_j} \subseteq U_{\tilde{n}}$, but $y(\tilde{n}) = 0$ and $y(m_j) = 1$ contradicts $(\dagger)$. Thus $U_n \cap F \neq \varnothing$ since $x \in U_n \cap F$. $\square$

Example 5.9 (Application)

Recall that every Polish space is homeomorphic to a closed subset of $\mathbb{R}^{\mathbb{N}}$, so it can be regarded as a point in $\mathcal{F}(\mathbb{R}^{\mathbb{N}})$. Thus we can regard $\mathcal{F}(\mathbb{R}^{\mathbb{N}})$ as the standard Borel space of all Polish spaces, and consider homeomorphism between Polish spaces as an equivalence relation on $\mathcal{F}(\mathbb{R}^{\mathbb{N}})$. But it is not an analytic equivalence relation, i.e. it is not analytic as a subset of $\mathcal{F}(\mathbb{R}^{\mathbb{N}})^2$; see Corollary 7.18 in §7.3. It is, however, $\mathbf{\Sigma}^1_2$; see Proposition 7.22. □

Definition 5.10

We call a subset of a standard Borel space $(X, \mathcal{S})$ analytic if it is analytic ($\mathbf{\Sigma}^1_1$, Definition 3.1) with respect to some (equivalently, every) compatible Polish topology, i.e. a Polish topology $\mathcal{T}$ on X with $\mathcal{B}(\mathcal{T}) = \mathcal{S}$: if $\mathcal{T}, \mathcal{T}'$ are both compatible Polish topologies for X, then $\mathrm{id} : (X, \mathcal{T}) \to (X, \mathcal{T}')$ is a Borel isomorphism, and the class $\mathbf{\Sigma}^1_1$ is closed under Borel images and preimages. □

We can also select points from closed sets in a Borel fashion.

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Definition 5.11

For a Polish space X, call a map $s : \mathcal{F}(X)^* \to X$, where $\mathcal{F}(X)^* := \mathcal{F}(X) \setminus \{\varnothing\}$, a selector if $s(F) \in F$ for each $F \in \mathcal{F}(X)^*$. We say that s is a Borel selector if it is a Borel map. □

Example 5.12

(a) For $X = \mathbb{R}$, the following s is a Borel selector: $s : \mathcal{F}(\mathbb{R})^* \to \mathbb{R}$ by

$$F \mapsto \min\left([-M, M] \cap F\right), \quad \text{where } M := \inf\{N > 0 : [-N, N] \cap F \neq \varnothing\},$$

or

$$F \mapsto \begin{cases} \min\{d \geq 0 : d \in F\}, \\ \max\{d < 0 : d \in F\} \quad \text{if the min does not exist.} \end{cases}$$

It is an exercise to check that this map is well-defined and Borel.

(b) For $X := \mathbb{N}^{\mathbb{N}}$, the following is a Borel selector: let $s : \mathcal{F}(\mathbb{N}^{\mathbb{N}})^* \to \mathbb{N}^{\mathbb{N}}$ by $F \mapsto$ the leftmost branch in T_F, where $T_F := \{x|_n : x \in F\} \subseteq \mathbb{N}^{<\mathbb{N}}$ is the tree of F. By leftmost, we mean the $x \in [T_F]$ recursively defined by $x(n) :=$ the least $j \in \mathbb{N}$ with $(x|_n)^\frown j \in T_F$, where $x|_n$ is already defined by induction; it exists since T_F is pruned. □

Theorem 5.13 (Borel axiom of choice for closed sets)

For every Polish space X, the Effros Borel space $\mathcal{F}(X)$ admits a Borel selector.

Proof. By HW 2, Q10, there is a continuous open surjection $f : \mathbb{N}^{\mathbb{N}} \to X$, so $f^{-1}(F)$ is closed for each

$F \in \mathcal{F}(X)$. This gives a map $\tilde{f} : \mathcal{F}(X) \to \mathcal{F}(\mathbb{N}^{\mathbb{N}})$, which is Borel since f is open.

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Hence if $s_{\mathbb{N}^{\mathbb{N}}} : \mathcal{F}(\mathbb{N}^{\mathbb{N}})^* \to \mathbb{N}^{\mathbb{N}}$ is a Borel selector, then $f \circ s_{\mathbb{N}^{\mathbb{N}}} \circ \tilde{f} : \mathcal{F}(X)^* \to X$ is a Borel selector. The details are filled in HW 4, Q10. □

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6 Projective Hierarchy

We now define a hierarchy of sets in Polish spaces obtained from open sets by using the operations $\neg$ (complements), $\vee$ (union), $\wedge$ (intersection), $\exists^{\mathbb{N}^{\mathbb{N}}}$ (projection along an $\mathbb{N}^{\mathbb{N}}$ coordinate) and $\forall^{\mathbb{N}^{\mathbb{N}}}$.

6.1 The Projective Classes

Definition 6.1 (Projective classes)

For each $n \geq 1$, define the projective classes inductively as follows: $\mathbf{\Sigma}^1_1$ is the class of analytic sets (as before), and supposing $\mathbf{\Sigma}^1_n$ has been defined,

$$\mathbf{\Pi}^1_n := \neg \mathbf{\Sigma}^1_n, \qquad \mathbf{\Sigma}^1_{n+1} := \exists^{\mathbb{N}^{\mathbb{N}}} \mathbf{\Pi}^1_n, \qquad \mathbf{\Delta}^1_n := \mathbf{\Sigma}^1_n \wedge \mathbf{\Pi}^1_n. \quad \square$$

Proposition 6.2 (Universal closure properties)

All classes $\mathbf{\Sigma}^1_n$, $\mathbf{\Pi}^1_n$, $\mathbf{\Delta}^1_n$ are closed under countable unions, countable intersections, and Borel preimages.

Proof. Induction on n, with $P(n) :=$ "everything above holds for $\mathbf{\Sigma}^1_n$ and $\mathbf{\Pi}^1_n$".

Base case ($n = 1$): Proposition 3.4.

Countable $\cup$: Suppose $P(n)$. Let $A_k = \mathrm{proj}_X(B_k)$, with $A_k \in \mathbf{\Sigma}^1_{n+1}(X)$, $B_k \in \mathbf{\Pi}^1_n(X \times \mathbb{N}^{\mathbb{N}})$. Then since maps commute with unions, $\bigcup_k A_k = \mathrm{proj}_X\left(\bigcup_k B_k\right)$, where $\bigcup_k B_k \in \mathbf{\Pi}^1_n(X \times \mathbb{N}^{\mathbb{N}})$ by the inductive hypothesis. Hence $\mathbf{\Sigma}^1_{n+1}(X)$ is closed under countable unions.

Countable $\cap$: Notice that

$$\begin{aligned} x \in \bigcap_{k \in \mathbb{N}} A_k &\iff \forall k \in \mathbb{N},\ x \in \mathrm{proj}_X B_k \\ &\iff \forall k \in \mathbb{N},\ \exists n_k \in \mathbb{N}^{\mathbb{N}} \text{ with } (x, n_k) \in B_k \\ &\iff \exists y \in (\mathbb{N}^{\mathbb{N}})^{\mathbb{N}}\ \forall k \in \mathbb{N}\ (x, y(k)) \in B_k. \end{aligned}$$

Then observe that, with $\tilde{B}_k := \{(x, y) \in X \times (\mathbb{N}^{\mathbb{N}})^{\mathbb{N}} : (x, y(k)) \in B_k\}$, $\tilde{B}_k$ is the continuous preimage of B_k by the map

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$\pi_k : X \times (\mathbb{N}^{\mathbb{N}})^{\mathbb{N}} \to X \times \mathbb{N}^{\mathbb{N}}$, $(x, y) \mapsto (x, y(k))$. Hence by the induction hypothesis ($\mathbf{\Pi}^1_n$ is closed under Borel preimages), $\tilde{B}_k = \pi_k^{-1}(B_k)$ is $\mathbf{\Pi}^1_n(X \times (\mathbb{N}^{\mathbb{N}})^{\mathbb{N}})$. Thus $\bigcap_k A_k = \operatorname{proj}_X\left(\bigcap_k \tilde{B}_k\right)$ is in $\exists^{(\mathbb{N}^{\mathbb{N}})^{\mathbb{N}}} \mathbf{\Pi}^1_n(X)$ by definition, which is $\exists^{\mathbb{N}^{\mathbb{N}}} \mathbf{\Pi}^1_n(X) = \mathbf{\Sigma}^1_{n+1}(X)$ by the homeomorphism $(\mathbb{N}^{\mathbb{N}})^{\mathbb{N}} \cong \mathbb{N}^{\mathbb{N}}$. $\mathbf{\Pi}^1_{n+1}$ is closed under countable $\cup$ and countable $\cap$ follows by taking complements with $\mathbf{\Sigma}^1_{n+1}$.

Borel preimages: Suppose $f : X \to Y$ is Borel with $A \in \mathbf{\Sigma}^1_{n+1}(Y)$. Then $A = \operatorname{proj}_Y(B)$ for some $B \in \mathbf{\Pi}^1_n(Y \times \mathbb{N}^{\mathbb{N}})$. Define $\tilde{f} : X \times \mathbb{N}^{\mathbb{N}} \to Y \times \mathbb{N}^{\mathbb{N}}$ by $(x, n) \mapsto (f(x), n)$, which is Borel since f is Borel. Then

$$f^{-1}(A) = f^{-1}(\operatorname{proj}_Y(B)) = \operatorname{proj}_X(\tilde{f}^{-1}(B)).$$

But $\mathbf{\Pi}^1_n$ is closed under Borel preimages by inductive hypothesis, hence $f^{-1}(A) \in \exists^{\mathbb{N}^{\mathbb{N}}} \mathbf{\Pi}^1_n(X) = \mathbf{\Sigma}^1_{n+1}(X)$. $\mathbf{\Pi}^1_{n+1}$ follows by taking complements. All statements for $\mathbf{\Delta}^1_{n+1}$ follow by taking intersections. □

Proposition 6.3

For Polish spaces, $\mathbf{\Sigma}^1_n \subseteq \mathbf{\Sigma}^1_{n+1}$ and $\mathbf{\Sigma}^1_n \subseteq \mathbf{\Pi}^1_{n+1}$. As a consequence, $\mathbf{\Sigma}^1_n \cup \mathbf{\Pi}^1_n \subseteq \mathbf{\Delta}^1_{n+1}$.

Proof. We prove $P(n)$: for all Polish spaces Y, $\mathbf{\Sigma}^1_n(Y) \subseteq \mathbf{\Sigma}^1_{n+1}(Y)$.

Base case: By the equivalent definitions of analytic sets, $A = \operatorname{proj}_X(F)$ for some closed $F \subseteq X \times \mathbb{N}^{\mathbb{N}}$. Closed sets are Borel, hence $\mathbf{\Delta}^1_1$. Thus $F \in \mathbf{\Pi}^1_1(X \times \mathbb{N}^{\mathbb{N}})$, hence $A \in \exists^{\mathbb{N}^{\mathbb{N}}} \mathbf{\Pi}^1_1(X) = \mathbf{\Sigma}^1_2(X)$.

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Inductive step: Taking complements, $\mathbf{\Pi}^1_n \subseteq \mathbf{\Pi}^1_{n+1}$. Now for $A \in \mathbf{\Sigma}^1_{n+1}(X)$, $\exists\, B \in \mathbf{\Pi}^1_n(X \times \mathbb{N}^{\mathbb{N}})$ with $A = \operatorname{proj}_X(B)$. Apply the hypothesis to $Y = X \times \mathbb{N}^{\mathbb{N}}$: $B \in \mathbf{\Pi}^1_{n+1}(X \times \mathbb{N}^{\mathbb{N}})$. Hence $A \in \exists^{\mathbb{N}^{\mathbb{N}}} \mathbf{\Pi}^1_{n+1}(X) = \mathbf{\Sigma}^1_{n+2}(X)$.

Now for $B \in \mathbf{\Pi}^1_n(X)$, $B \times \mathbb{N}^{\mathbb{N}} \in \mathbf{\Pi}^1_n(X \times \mathbb{N}^{\mathbb{N}})$ since $B \times \mathbb{N}^{\mathbb{N}} = \operatorname{proj}_X^{-1}(B)$ and $\mathbf{\Pi}^1_n$ is closed under Borel preimages, hence $B \in \exists^{\mathbb{N}^{\mathbb{N}}} \mathbf{\Pi}^1_n(X) = \mathbf{\Sigma}^1_{n+1}(X)$ by definition; equivalently, taking complements, $\mathbf{\Sigma}^1_n \subseteq \mathbf{\Pi}^1_{n+1}$. □

Corollary 6.4 (The projective hierarchy)

For a Polish space X,

$$\mathbf{P}(X) := \bigcup_{n \in \mathbb{N}^+} \mathbf{\Sigma}^1_n(X) = \bigcup_{n \in \mathbb{N}^+} \mathbf{\Pi}^1_n(X) = \bigcup_{n \in \mathbb{N}^+} \mathbf{\Delta}^1_n(X).$$

We call the sets in $\mathbf{P}$ projective, and $\mathbf{P}$ the projective class. We also have the following projective hierarchy:

$$\begin{array}{ccccccccc} & & \mathbf{\Sigma}^1_1(X) & & & & \mathbf{\Sigma}^1_n(X) & & \\ & \nearrow & & \searrow & & \nearrow & & \searrow & \\ \mathcal{B}(X) = \mathbf{\Delta}^1_1(X) & & & & \mathbf{\Delta}^1_2(X) \cdots & \mathbf{\Delta}^1_n(X) & & & \mathbf{\Delta}^1_{n+1}(X) \cdots \\ & \searrow & & \nearrow & & \searrow & & \nearrow & \\ & & \mathbf{\Pi}^1_1(X) & & & & \mathbf{\Pi}^1_n(X) & & \end{array}$$

Proposition 6.5 (Closure properties)

For Polish spaces X,

(a) $\mathbf{\Sigma}^1_n$ is closed under Borel images.
(b) $\mathbf{\Pi}^1_n$ is closed under coprojections, i.e. for any Polish space Y, $\forall^Y \mathbf{\Pi}^1_n \subseteq \mathbf{\Pi}^1_n$.
(c) $\mathbf{\Delta}^1_n$ is closed under complements, hence it is a σ-algebra.

Proof. (a) $n = 1$ is already proven. Now $\mathbf{\Sigma}^1_{n+1} = \exists^{\mathbb{N}^{\mathbb{N}}} \mathbf{\Pi}^1_n$; let $f : X \to Y$ be Borel between Polish spaces, and $A \in \mathbf{\Sigma}^1_{n+1}(X)$, so $A = \mathrm{proj}_X(B)$ with $B \in \mathbf{\Pi}^1_n(X \times \mathbb{N}^{\mathbb{N}})$. Let $g : \mathbb{N}^{\mathbb{N}} \twoheadrightarrow X \times \mathbb{N}^{\mathbb{N}}$ be a continuous surjection. Then for each $y \in Y$, we have

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$$\begin{aligned} y \in f(A) &\iff \exists\, x \in X \ \big(x \in A \text{ and } f(x) = y\big) \\ &\iff \exists\, x \in X\ \exists\, z \in \mathbb{N}^{\mathbb{N}} \ \big((x,z) \in B \text{ and } f(x) = y\big) \\ &\iff \exists\, w \in \mathbb{N}^{\mathbb{N}} \text{ s.t. } g(w) \in B \ \wedge\ f\big(\mathrm{proj}_X(g(w))\big) = y \\ &\iff \underbrace{\exists\, w \in \mathbb{N}^{\mathbb{N}}}_{\substack{\exists^{\mathbb{N}^{\mathbb{N}}},\\ \text{i.e. } \mathrm{proj}_Y}} \text{ s.t. } \underbrace{w \in g^{-1}(B)}_{\substack{\mathbf{\Pi}^1_n, \text{ since}\\ g \text{ is continuous}}} \wedge \underbrace{(w,y) \in \mathrm{graph}(f \circ \mathrm{proj}_X \circ g)}_{\substack{\mathbf{\Delta}^1_1 = \text{Borel, since}\\ f \circ \mathrm{proj}_X \circ g \text{ is Borel}}}. \end{aligned}$$

Hence $f(A) \in \exists^{\mathbb{N}^{\mathbb{N}}} \mathbf{\Pi}^1_n = \mathbf{\Sigma}^1_{n+1}$.

(b) $\forall^Y \mathbf{\Pi}^1_n = \neg\exists^Y \neg\mathbf{\Pi}^1_n = \neg\exists^Y \mathbf{\Sigma}^1_n \subseteq \neg\mathbf{\Sigma}^1_n = \mathbf{\Pi}^1_n$, where $\exists^Y \mathbf{\Sigma}^1_n \subseteq \mathbf{\Sigma}^1_n$ since $A = \mathrm{proj}_X(B)$, $B \in \mathbf{\Sigma}^1_n(X \times Y)$, and $\mathbf{\Sigma}^1_n$ is closed under Borel images.

(c) $\mathbf{\Delta}^1_n$ is closed by definition, and together with the universal closure properties, it is a σ-algebra. □

Corollary 6.6

For any uncountable Polish space Y and $n \geq 1$, $\mathbf{\Sigma}^1_{n+1} = \exists^Y \mathbf{\Pi}^1_n$ and $\mathbf{\Pi}^1_{n+1} = \forall^Y \mathbf{\Sigma}^1_n$.

Proof. By the Borel Isomorphism Theorem, $\exists\, f : \mathbb{N}^{\mathbb{N}} \to Y$ Borel isomorphism, thus $\tilde{f} : X \times \mathbb{N}^{\mathbb{N}} \to X \times Y$ by $(x, n) \mapsto (x, f(n))$ is a Borel isomorphism. Then it follows from the fact that $\mathbf{\Pi}^1_n$ is closed under Borel preimages that $\mathbf{\Sigma}^1_{n+1} = \exists^{\mathbb{N}^{\mathbb{N}}} \mathbf{\Pi}^1_n = \exists^Y \mathbf{\Pi}^1_n$. Taking the complement we get

$$\mathbf{\Pi}^1_{n+1} = \neg\mathbf{\Sigma}^1_{n+1} = \neg\exists^Y \mathbf{\Pi}^1_n = \neg\exists^Y \neg\mathbf{\Sigma}^1_n = \forall^Y \mathbf{\Sigma}^1_n.$$ □

6.2 Strictness of the Projective Hierarchy

Proposition 6.7

For any Polish spaces X, Y, if X is uncountable, then for each $n \in \mathbb{N}^+$, there exists an X-universal set $U \subseteq$

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$X \times Y$ for $\mathbf{\Sigma}^1_n(Y)$, i.e. $\{U_x : x \in X\} = \mathbf{\Sigma}^1_n(Y)$ and $U \in \mathbf{\Sigma}^1_n(X \times Y)$. Consequently, there is also an X-universal set for $\mathbf{\Pi}^1_n(Y)$.

Proof. By induction on $n \in \mathbb{N}^+$, with the base case already proven. Now to get an X-universal set, set $F \subseteq X \times Y \times \mathbb{N}^{\mathbb{N}}$ an X-universal set for $\mathbf{\Pi}^1_n(Y \times \mathbb{N}^{\mathbb{N}})$. Then verify $U := \mathrm{proj}_{X\times Y}(F)$ is an X-universal set for $\mathbf{\Sigma}^1_{n+1}(Y)$ from the definition and the fact that fibers commute with projections. One verifies from the definition that U^c is an X-universal set for $\mathbf{\Pi}^1_{n+1}(Y)$. □

Corollary 6.8 (Strictness of the projective hierarchy)

The projective hierarchy is strict in every uncountable Polish space X, i.e.

$$\mathbf{\Sigma}^1_n(X) \subsetneq \mathbf{\Delta}^1_{n+1}(X) \subsetneq \mathbf{\Sigma}^1_{n+1}(X) \qquad \text{and} \qquad \mathbf{\Pi}^1_n(X) \subsetneq \mathbf{\Delta}^1_{n+1}(X) \subsetneq \mathbf{\Pi}^1_{n+1}(X).$$

Proof. Let $U \subseteq X \times X$ be an X-universal set for $\mathbf{\Sigma}^1_n(X)$, so $A := \mathrm{AntiDiag}(U) = \{x \in X : (x, x) \notin U\}$ is $\mathbf{\Pi}^1_n(X)$, being the Borel preimage of U^c under the map $x \mapsto (x, x)$. By Cantor's diagonalization, A is not a fiber of U, hence $A \notin \mathbf{\Sigma}^1_n(X)$. Thus $A \in \mathbf{\Delta}^1_{n+1}(X) \setminus \mathbf{\Sigma}^1_n(X)$ and $A^c \in \mathbf{\Delta}^1_{n+1}(X) \setminus \mathbf{\Pi}^1_n(X)$, which gives the first inclusion of each chain; the same sets built for $n+1$ in place of n lie in $\mathbf{\Sigma}^1_{n+1}(X) \setminus \mathbf{\Delta}^1_{n+1}(X)$ and $\mathbf{\Pi}^1_{n+1}(X) \setminus \mathbf{\Delta}^1_{n+1}(X)$, giving the second. □

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7 Γ-Complete Sets

7.1 Wadge Reducibility

Definition 7.1

Let X, Y be sets and $A \subseteq X$, $B \subseteq Y$. A map $f : X \to Y$ is called a reduction of A to B if $f^{-1}(B) = A$, i.e. $\forall x \in X,\ x \in A \iff f(x) \in B$.

For topological spaces X, Y, we say that A is Wadge reducible to B, and write $A \leq_{\mathrm{W}} B$, if there is a continuous reduction $f : X \to Y$ of A to B. □

Definition 7.2

Let Y be a Polish space, Γ be a class of sets, e.g. $\mathbf{\Sigma}^1_7$. A set $B \subseteq Y$ is called Γ-hard if for each zero-dimensional Polish space X and each $A \in \Gamma(X)$, $A \leq_W B$. Moreover, if $B \in \Gamma(Y)$, then we say that B is Γ-complete. □

Observation 7.3

If Γ is a class of sets closed under continuous preimages, X is Polish and $B \in \Gamma(X)$, then $A \leq_W B \implies A \in \Gamma$. □

Corollary 7.4

If Γ is a class of sets closed under continuous preimages and $\Gamma(X) \neq \neg\Gamma(X)$ for some zero-dimensional Polish space X, then every Γ-hard set is not in $\neg\Gamma$.

Proof. $\neg\Gamma$ is also closed under continuous preimages. If B is Γ-hard and in $\neg\Gamma$, then for all $A \in \Gamma(X)$, $\exists\, f_A : X \to Y$ continuous with $f_A^{-1}(B) = A$, hence $A \in \Gamma \cap \neg\Gamma$. So $\Gamma(X) \subseteq \neg\Gamma(X)$, and taking complements, $\Gamma(X) = \neg\Gamma(X)$. But this is a contradiction since $\Gamma(X) \neq \neg\Gamma(X)$. □

For the Borel hierarchy, the converse holds:

Theorem 7.5 (Wadge)

Let Y be a zero-dimensional Polish space. Fix $\xi < \omega_1$. A Borel set $B \subseteq Y$ is $\mathbf{\Sigma}^0_\xi$-hard $\iff$ $B \notin \mathbf{\Pi}^0_\xi$.

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Thus B is $\mathbf{\Sigma}^0_\xi$-complete $\iff$ $B \in \mathbf{\Sigma}^0_\xi \setminus \mathbf{\Pi}^0_\xi$. The same result holds with $\mathbf{\Sigma}^0_\xi$ and $\mathbf{\Pi}^0_\xi$ swapped throughout.

Proof. Postponed, since it uses infinite games and Borel determinacy. □

7.2 The Space of Trees and Ill-Founded Trees

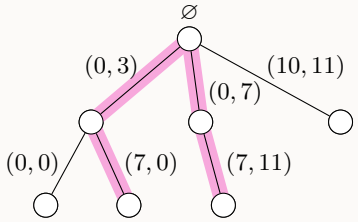
Example 7.6 (Analytic sets via trees)

Every analytic subset $A \subseteq \mathbb{N}^\mathbb{N}$ is a projection of a closed set $F \subseteq \mathbb{N}^\mathbb{N} \times \mathbb{N}^\mathbb{N}$. For convenience, identify $\mathbb{N}^\mathbb{N} \times \mathbb{N}^\mathbb{N} \cong (\mathbb{N} \times \mathbb{N})^\mathbb{N}$ by the natural homeomorphism $((x_n)_n, (y_n)_n) \mapsto (x_n, y_n)_n$. This turns the projection proj_0 into $p : (\mathbb{N} \times \mathbb{N})^\mathbb{N} \to \mathbb{N}^\mathbb{N}$ given by $(x_n, y_n)_n \mapsto (x_n)_n$. Closed subsets of $(\mathbb{N} \times \mathbb{N})^\mathbb{N}$ are exactly the sets $[T]$ for some pruned tree T on the set $\mathbb{N} \times \mathbb{N}$. Hence a set $A \subseteq \mathbb{N}^\mathbb{N}$ is analytic $\iff$ $A = p([T]) =: p[T]$, for some pruned tree T on $\mathbb{N} \times \mathbb{N}$.

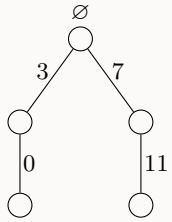
For a tree T on $\mathbb{N} \times \mathbb{N}$ and $x \in \mathbb{N}^\mathbb{N}$, define

$$T_x := \{s \in \mathbb{N}^{<\mathbb{N}} : (x|_{|s|}, s) \in T\};$$

e.g. for $x = 0\,7\,5\cdots$:



T



T_x

Then note that for each $x \in \mathbb{N}^{\mathbb{N}}$,

$$\begin{aligned} x \in p[T] &\iff \exists\, y \in \mathbb{N}^{\mathbb{N}} \text{ s.t. } (x_n, y_n)_n \in [T] \\ &\iff \exists\, y \in \mathbb{N}^{\mathbb{N}} \text{ s.t. } y|_n \in T_x,\ \forall n \in \mathbb{N} \\ &\iff \exists\, y \in \mathbb{N}^{\mathbb{N}} \text{ s.t. } y \in [T_x] \\ &\iff [T_x] \neq \varnothing. \end{aligned} \tag{$*$}$$

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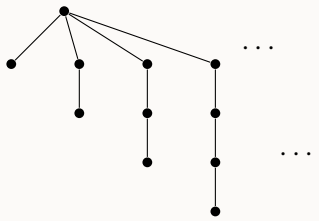
$(*)$ gives a way of defining a complete analytic set. □

Definition 7.7

A tree T on a set A (i.e. $T \subseteq A^{<\mathbb{N}}$) is ill-founded if $[T] \neq \varnothing$; otherwise call it well-founded. □

Example 7.8

The tree



is a well-founded infinite tree. □

Definition 7.9 (The space of trees)

To build the Polish space of all trees on $\mathbb{N}$, we identify $\mathcal{P}(\mathbb{N}^{<\mathbb{N}})$ with the Cantor space $2^{\mathbb{N}^{<\mathbb{N}}} \cong 2^{\mathbb{N}}$, since $\mathbb{N}^{<\mathbb{N}}$ is countable. Define

$$\begin{aligned} \mathrm{Tr} &:= \{T \in \mathcal{P}(\mathbb{N}^{<\mathbb{N}}) : T \text{ is a tree}\} \\ &= \{T \in \mathcal{P}(\mathbb{N}^{<\mathbb{N}}) : \varnothing \in T \text{ and } \forall s,t \in \mathbb{N}^{<\mathbb{N}} \text{ with } s \subseteq t\ [t \in T \implies s \in T]\} \subseteq \mathcal{P}(\mathbb{N}^{<\mathbb{N}}), \end{aligned}$$

the space of trees on $\mathbb{N}$, which is closed in $\mathcal{P}(\mathbb{N}^{<\mathbb{N}}) \cong 2^{\mathbb{N}}$:

$$\mathrm{Tr} = \{S \in 2^{\mathbb{N}^{<\mathbb{N}}} : \varnothing \in S\} \cap \bigcap_{s \subseteq t} \{S \in 2^{\mathbb{N}^{<\mathbb{N}}} : t \notin S \ \vee \ s \in S\},$$

each of them is a finite membership condition for S, hence Tr is an intersection of clopen sets in $2^{\mathbb{N}^{<\mathbb{N}}}$, thus closed. □

Corollary 7.10

Tr is Polish.

Proof. Tr is a closed subset of the Polish space $2^{\mathbb{N}^{<\mathbb{N}}}$

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$\cong 2^{\mathbb{N}}$. □

Definition 7.11

Denote the class of ill-founded trees on $\mathbb{N}$ by IL (or IF), i.e.

$$\mathrm{IL} := \{T \in \mathrm{Tr} : [T] \neq \varnothing\} \subseteq \mathrm{Tr}. \quad \square$$

Observation 7.12 (IL is analytic)

For each $T \in \mathrm{Tr}$,

$$T \in \mathrm{IL} \iff \underbrace{\exists x \in \mathbb{N}^{\mathbb{N}} \ \underbrace{\forall n \in \mathbb{N} \quad \underbrace{[x|_n \in T]}_{\text{closed subset of } \mathrm{Tr}\times\mathbb{N}^{\mathbb{N}}}}_{\text{closed}}}_{\text{analytic}}.$$

Explicitly, $\mathrm{IL} = \mathrm{proj}_{\mathrm{Tr}}\left(\bigcap_{n\in\mathbb{N}} C_n\right)$, where $C_n := \{(T, x) \in \mathrm{Tr} \times \mathbb{N}^{\mathbb{N}} : x|_n \in T\}$ is closed, since for every $(T, x) \notin C_n$, $\{T' \in \mathrm{Tr} : x|_n \notin T'\} \times [x|_n]_{\mathbb{N}}$ is an open set in $\mathrm{Tr} \times \mathbb{N}^{\mathbb{N}}$ containing (T, x) and disjoint from C_n. □

Theorem 7.13

IL is $\mathbf{\Sigma}^1_1$-complete.

Proof. Since every zero-dimensional Polish space X is homeomorphic to a closed subset of $\mathbb{N}^{\mathbb{N}}$, analytic subsets of X would be analytic subsets of $\mathbb{N}^{\mathbb{N}}$ (for instance, by (3) of the equivalent definitions of analyticity), hence it is enough to prove for $X = \mathbb{N}^{\mathbb{N}}$.

Let $A \subseteq \mathbb{N}^{\mathbb{N}}$ be analytic, hence $\exists T$ tree on $\mathbb{N} \times \mathbb{N}$ s.t. $A = p[T]$. Then the map $f : \mathbb{N}^{\mathbb{N}} \to \mathrm{Tr}$

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by $x \mapsto T_x$ is a Wadge reduction of A to IL: f is a reduction since $\forall x \in \mathbb{N}^{\mathbb{N}}$, $x \in A = p[T] \iff [T_x] \neq \varnothing \iff T_x \in \mathrm{IL}$. f is continuous since the sets

$$[(s)] := \{T' \in \mathrm{Tr} : s \in T'\} = \{T' \in 2^{\mathbb{N}^{<\mathbb{N}}} : s \in T'\} \cap \mathrm{Tr}, \qquad s \in \mathbb{N}^{<\mathbb{N}},$$

together with their complements form a subbasis of Tr, and $f^{-1}([(s)]) = \{x \in \mathbb{N}^{\mathbb{N}} : (x|_{|s|}, s) \in T\}$ is clopen. Lastly $\mathrm{IL} \in \mathbf{\Sigma}^1_1$ by the previous observation. □

Corollary 7.14

The set WF := Tr \ IL of well-founded trees is $\mathbf{\Pi}^1_1$-complete.

Proof. Take complements on IL in the previous theorem. □

7.3 Borel Reducibility and the Homeomorphism Relation

Definition 7.15

For topological spaces X, Y and $A \subseteq X$, $B \subseteq Y$, we say that A is Borel reducible to B if there is a reduction $f : X \to Y$ of A to B that is Borel, and we write $A \leq_{\mathrm{B}} B$. □

Definition 7.16

Let Y be a Polish space, $B \subseteq Y$, and Γ a class of sets. We say that B is Γ-Borel-hard if for every zero-dimensional Polish space X and every $A \in \Gamma(X)$, $A \leq_{\mathrm{B}} B$. If moreover $B \in \Gamma(Y)$, then we say that B is Γ-Borel-complete. □

Observation 7.17

(a) For X Polish, if $A \subseteq X$ is Γ-complete (resp. Γ-hard), then A is Γ-Borel-complete (resp. Γ-Borel-hard).

(b) If Γ is a class closed under Borel preimages, X is Polish and $B \in \Gamma(X)$, then $A \leq_{\mathrm{B}} B \implies A \in \Gamma$.

(c) If Γ is a class of sets closed under

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Borel preimages and $\Gamma(X) \neq \neg\Gamma(X)$ for some zero-dimensional Polish space X, then every Γ-Borel-hard set is not in $\neg\Gamma$.

Proof. Same as for Γ-Wadge-complete and Γ-Wadge-hard sets. □

Corollary 7.18 (Homeomorphism is not analytic)

Recall that the collection of Polish spaces can be identified with the Effros space $\mathcal{F}(\mathbb{R}^{\mathbb{N}})$, and homeomorphism can be viewed as an equivalence relation on $\mathcal{F}(\mathbb{R}^{\mathbb{N}})$, i.e. $E \subseteq \mathcal{F}(\mathbb{R}^{\mathbb{N}})^2$. Then E is not analytic, i.e. $E \notin \mathbf{\Sigma}^1_1$.

Proof. We first show that $[\mathbb{N}]_{\cong} := \{F \in \mathcal{F}(\mathbb{R}^{\mathbb{N}}) : F \cong \mathbb{N}\}$ is $\mathbf{\Pi}^1_1$-Borel-complete. Let W be a tree on $\mathbb{N}$, and $t = (t_0, t_1, \dots, t_{k-1}) \in W$. Define

$$\begin{aligned} x_t &:= (t_0, \dots, t_{k-1}, -1, -1, \dots) \in \mathbb{R}^{\mathbb{N}}, \\ X_W &:= \overline{\{x_t : t \in W\}}. \end{aligned}$$

A sequence $(x_{t_j})_j$ converges $\iff$ it is eventually constant in each coordinate with respect to the metric on $\mathbb{R}^{\mathbb{N}}$, i.e. $\forall n \in \mathbb{N}$, $(x_{t_j}(n))_j$ eventually becomes a constant. From this, one can check that

(1) $X_W = \{x_t : t \in W\} \cup [W]$;
(2) x_t is isolated in X_W, i.e. $\{x_t\}$ is open;
(3) each $b \in [W]$ is the limit of $(x_{b|_k})_{k\in\mathbb{N}}$, hence not isolated.

So X_W is discrete $\iff$ $[W] = \varnothing$ $\iff$ $W \in \mathrm{WF}$.

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The map $\mathrm{Tr} \to \mathcal{F}(\mathbb{R}^{\mathbb{N}})$ by $W \mapsto X_W$ is Borel, because for any generating Borel set $[U] = \{F \in \mathcal{F}(\mathbb{R}^{\mathbb{N}}) : F \cap U \neq \varnothing\}$, $X_W \cap U \neq \varnothing$ $\iff$ $x_t \in U$ for some $t \in W$, by (1) and (3). Replacing W by

$$W' := \{\varnothing\} \cup \{(n)^\frown t : n \in \mathbb{N},\ t \in W\}$$

makes W infinite without changing well-foundedness of W. Hence

$$W \in \mathrm{WF} \iff W' \in \mathrm{WF} \iff X_{W'} \text{ is discrete} \iff X_{W'} \cong \mathbb{N},$$

with the last equivalence because $X_{W'}$ is countably infinite and discrete. Therefore the set WF of well-founded trees on $\mathbb{N}$ is Borel reducible to $[\mathbb{N}]_{\cong}$. By the last corollary, WF is $\mathbf{\Pi}^1_1$-complete, hence $\mathbf{\Pi}^1_1$-Borel-complete by (a) of the observation above, hence $[\mathbb{N}]_{\cong}$ is $\mathbf{\Pi}^1_1$-Borel-hard.

Moreover, $[\mathbb{N}]_{\cong} \in \mathbf{\Pi}^1_1$: fix a countable basis $(U_m)_m$ of $\mathbb{R}^{\mathbb{N}}$, then observe that $F \cong \mathbb{N}$ $\iff$

$$\underbrace{\forall n \in \mathbb{N}\ \exists m_1, \dots, m_n \left[\forall i < j\ (U_{m_i} \cap U_{m_j} = \varnothing)\ \wedge\ \forall i \leq n\ (F \cap U_{m_i} \neq \varnothing)\right]}_{\text{(I)}}$$

$$\wedge\ \forall x \in \mathbb{R}^{\mathbb{N}} \Big[\underbrace{\forall k\ (x \in U_k \implies F \cap U_k \neq \varnothing)}_{\text{(II)}}$$

$$\implies \underbrace{\exists m\ (x \in U_m\ \wedge\ \forall m'\ (U_{m'} \subseteq U_m\ \wedge\ x \notin U_{m'} \implies F \cap U_{m'} = \varnothing))}_{\text{(III)}}\Big].$$

(I) says $|F| \geq n$ for all $n \in \mathbb{N}$, so F is infinite. (II): F is closed, so every open basic neighbourhood

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of x meets F $\iff$ $x \in F$. (II) $\Rightarrow$ (III) says that if $x \in F$, then F has no points in U_m other than x, i.e. x is isolated in F. (I) is countable operations applied to open/closed conditions on x and the Effros Borel generators $F \cap U \neq \varnothing$ (and the condition $U_{m'} \subseteq U_m$ on indices, which is nothing), hence Borel in $\mathcal{F}(\mathbb{R}^{\mathbb{N}})$. (II) and (III) are likewise open/closed conditions on x and Effros generators, hence Borel. The complement of the second clause is $\exists x \in \mathbb{R}^{\mathbb{N}}$ [(II) $\wedge \neg$(III)], which is the image of a Borel set in $\mathcal{F}(\mathbb{R}^{\mathbb{N}}) \times \mathbb{R}^{\mathbb{N}}$ (namely [(II) $\wedge \neg$(III)]) under the projection $\mathrm{proj}_0 : \mathcal{F}(\mathbb{R}^{\mathbb{N}}) \times \mathbb{R}^{\mathbb{N}} \to \mathcal{F}(\mathbb{R}^{\mathbb{N}})$, $(F, x) \mapsto F$,

hence analytic by one of the equivalent conditions. Thus $\forall x \in \mathbb{R}^{\mathbb{N}}\ [(\mathrm{II}) \Rightarrow (\mathrm{III})]$ is the complement of an analytic set, i.e. $\mathbf{\Pi}^1_1$. Therefore

$$[\mathbb{N}]_{\cong} = \underbrace{(\mathrm{I})}_{\text{Borel}} \cap \underbrace{\forall x \in \mathbb{R}^{\mathbb{N}}\ [(\mathrm{II}) \Rightarrow (\mathrm{III})]}_{\text{coanalytic}} \in \mathbf{\Pi}^1_1.$$

If $E \subseteq Z^2$ is analytic, then every class $[z_0]_E$ is the preimage of E under the Borel map $z \mapsto (z, z_0)$, hence analytic. But the homeomorphism class of $\mathbb{N}$ is $\mathbf{\Pi}^1_1$-Borel-complete, hence not analytic by (c) of the observation above. □

Definition 7.19

Let X be a Polish space with a fixed countable basis $(U_m)_m$. For F closed and $(x_n)_n$ a sequence in X, we write $\mathrm{Dense}(F, (x_n))$ if

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$$\forall n\ (x_n \in F)\ \wedge\ \forall m\ (F \cap U_m \neq \varnothing\ \Longrightarrow\ \exists n\ (x_n \in U_m)),$$

i.e. $(x_n)_n$ is dense in F with its relative topology. □

Lemma 7.20 (Extension lemma)

Let $F_1, F_2 \subseteq \mathbb{R}^{\mathbb{N}}$ be closed, with $\mathrm{Dense}(F_1, (x_n))$ and $\mathrm{Dense}(F_2, (y_n))$, and let d be a complete compatible metric on $\mathbb{R}^{\mathbb{N}}$. The following are equivalent:

(a) there is a homeomorphism $h : F_1 \xrightarrow{\sim} F_2$ with $h(x_n) = y_n$ for all n;
(b) for all $\alpha \in \mathbb{N}^{\mathbb{N}}$: $(x_{\alpha(k)})_k$ is Cauchy iff $(y_{\alpha(k)})_k$ is Cauchy.

Proof. ($\Rightarrow$) A Cauchy sequence converges in $\mathbb{R}^{\mathbb{N}}$, with limit in F_1 since F_1 is closed. Convergence is a topological property, hence $(h(x_{\alpha(k)}))_k = (y_{\alpha(k)})_k$ is convergent, thus Cauchy. The converse of (b) follows by the same argument with h^{-1}.

($\Leftarrow$) Applying (b) to the alternating sequence $x_n, x_m, x_n, x_m, \ldots$, we have

$$x_n = x_m \iff (x_n, x_m, x_n, x_m, \ldots) \text{ is Cauchy} \iff (y_n, y_m, y_n, y_m, \ldots) \text{ is Cauchy} \iff y_n = y_m.$$

So $g(x_n) := y_n$ is a well-defined bijection between $D_1 = \{x_n\}_n$ and $D_2 = \{y_n\}_n$. For $x \in F_1$, choose $x_{n_k} \to x$; then $(x_{n_k})_k$ is Cauchy, so $(y_{n_k})_k$ is Cauchy and converges in $\mathbb{R}^{\mathbb{N}}$ to a point of the closed set F_2, call it $h(x)$. Interleaving two sequences converging to x shows that h does not depend on the choice of $(x_{n_k})_k$. Constant sequences show that h extends g. If $z_j \to z \in F_1$, then choose n_j with $d(x_{n_j}, z_j) < 1/j$ and $d(y_{n_j}, h(z_j)) < 1/j$. So $x_{n_j} \to z$, so $y_{n_j} \to h(z)$

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by the definition of h, so $h(z_j) \to h(z)$. Symmetrically, g^{-1} extends to a continuous $h' : F_2 \to F_1$. Now $h' \circ h$ is the identity on the dense set D_1, hence on F_1. Likewise $h \circ h' = \mathrm{id}$ on D_2, hence on F_2. So h is a homeomorphism with inverse h'. □

Proposition 7.21 (Homeomorphism of compact spaces is analytic)

The equivalence relation of homeomorphism $K_1 \cong K_2$ on $\mathcal{F}([0,1]^{\mathbb{N}})$ is $\mathbf{\Sigma}^1_1$. Recall that compact Polish spaces are closed subsets of $[0,1]^{\mathbb{N}}$, so homeomorphism of compact Polish spaces is analytic in $\mathcal{F}([0,1]^{\mathbb{N}})$, hence analytic in $\mathcal{F}(\mathbb{R}^{\mathbb{N}})$ since the canonical injection $\mathcal{F}([0,1]^{\mathbb{N}}) \hookrightarrow \mathcal{F}(\mathbb{R}^{\mathbb{N}})$ is Borel.

Proof. Observe that for nonempty K_1, K_2 (the empty set is homeomorphic only to itself, a Borel condition), $K_1 \cong K_2 \iff \exists\, (x_n)_n, (y_n)_n \subseteq [0,1]^{\mathbb{N}}$ s.t. Dense$(K_1, (x_n)_n)$, Dense$(K_2, (y_n)_n)$, and

$$(*) \qquad \begin{aligned} &\forall \varepsilon \in \mathbb{Q}^+ \,\exists\, \delta \in \mathbb{Q}^+ \,\forall n, m \;\big(d(x_n, x_m) < \delta \implies d(y_n, y_m) < \varepsilon\big) \\ \wedge\; &\forall \varepsilon \in \mathbb{Q}^+ \,\exists\, \delta \in \mathbb{Q}^+ \,\forall n, m \;\big(d(y_n, y_m) < \delta \implies d(x_n, x_m) < \varepsilon\big). \end{aligned}$$

($\Rightarrow$) Dense$(K_1, (x_n))$ and Dense$(K_2, (y_n))$ follow by taking $(x_n)_n$ dense with $y_n := h(x_n)$, h the homeomorphism. $(*)$ follows from Heine–Cantor applied to h and h^{-1}.

($\Leftarrow$) follows from the Extension lemma and the fact that uniformly continuous maps send Cauchy sequences to Cauchy sequences.

Complexity: "Dense" is Borel in $\mathcal{F}([0,1]^{\mathbb{N}}) \times ([0,1]^{\mathbb{N}})^{\mathbb{N}}$,

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and $(*)$ is countable quantifiers over an open condition, also Borel. Hence the set of all tuples $(K_1, K_2, (x_n)_n, (y_n)_n)$ satisfying these is a Borel subset of $\mathcal{F}([0,1]^{\mathbb{N}})^2 \times (([0,1]^{\mathbb{N}})^{\mathbb{N}})^2$, and $\cong$ is its projection to $\mathcal{F}([0,1]^{\mathbb{N}})^2$, a projection of a Borel set, thus analytic by one of the equivalent conditions. □

Proposition 7.22 (Homeomorphism is $\mathbf{\Sigma}^1_2$)

The homeomorphism relation $F_1 \cong F_2$ on $\mathcal{F}(\mathbb{R}^{\mathbb{N}})$ is $\mathbf{\Sigma}^1_2$.

Proof. By the Extension lemma, for nonempty F_1, F_2, $F_1 \cong F_2$ iff $\exists\, (x_n), (y_n) \in (\mathbb{R}^{\mathbb{N}})^{\mathbb{N}}$ s.t.

$$\text{Dense}(F_1, (x_n)) \;\wedge\; \text{Dense}(F_2, (y_n)) \;\wedge\; \forall \alpha \in \mathbb{N}^{\mathbb{N}} \;\big(\text{Cauchy}(x_{\alpha(k)})_k \iff \text{Cauchy}(y_{\alpha(k)})_k\big).$$

Cauchy$(x_{\alpha(k)})_k$ is $\forall \varepsilon \in \mathbb{Q}^+ \,\exists N \,\forall k, l \geq N \;\big(d(x_{\alpha(k)}, x_{\alpha(l)}) < \varepsilon\big)$, a Borel condition on $(\alpha, (x_n)) \in \mathbb{N}^{\mathbb{N}} \times (\mathbb{R}^{\mathbb{N}})^{\mathbb{N}}$. Quantifying universally over a Polish space turns a Borel set into $\mathbf{\Pi}^1_1$. Hence, quantifying existentially over the Polish space $\big((\mathbb{R}^{\mathbb{N}})^{\mathbb{N}}\big)^2$, $\cong$ is $\exists^Y \mathbf{\Pi}^1_1 = \mathbf{\Sigma}^1_2$, as the projection may be taken along any uncountable Polish space. □